

Lévy-Score-Corrected Jump Diffusions for Accelerated Sampling of Boltzmann Distributions

Abstract

Sampling the Boltzmann distribution associated with an Itô diffusion process presents a fundamental challenge in computational stochastic simulation. Traditional numerical integration of the governing stochastic differential equations necessitates fine temporal discretization and prohibitively long simulation horizons, particularly in the presence of metastability. In this paper, we introduce a novel algorithmic framework employing a Lévy-score-based Monte Carlo method to directly generate samples from the target distribution. By leveraging the macroscopic jumps characteristic of Lévy processes, the proposed method efficiently facilitates rapid transitions across high-energy barriers. Crucially, this approach mitigates prohibitive computational costs while preserving the structural simplicity and algorithmic efficiency of standard Monte Carlo techniques. We rigorously establish the global well-posedness and geometric ergodicity of the jump-augmented stochastic differential equation, alongside a detailed theoretical analysis of the mechanisms driving the accelerated sampling rate. These theoretical guarantees ensure that the time-averaged trajectory converges robustly to the target distribution, thereby providing a strict mathematical justification for the proposed sampling algorithm. Numerical experiments across several benchmark problems demonstrate the efficacy and scalability of the approach. Compared to traditional diffusion-based sampling, our method achieves an efficiency improvement of several orders of magnitude. These results highlight its low computational overhead and superior capability in accelerating the exploration of complex potential landscapes characterized by multiple metastable states.

Keywords: Boltzmann distribution, Itô diffusion, Lévy process, Monte Carlo methods, stochastic simulation, metastable states

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1 Introduction

In statistical mechanics and computational physics, the *Boltzmann distribution* (also referred to as the *Gibbs distribution* [1]) serves as a fundamental probability measure. It characterizes the likelihood of a system occupying a given state x as a function of the state's energy and the system's temperature. The corresponding probability density function is expressed as

$$p(x) \propto \exp \left\{ -\frac{V(x)}{k_B \mathcal{T}} \right\}, \quad (1)$$

where $V(x)$ denotes the potential energy of the state x , k_B is the Boltzmann constant, and $\mathcal{T}$ is the absolute temperature of the system. The efficient computation and sampling of this Boltzmann distribution remains a central challenge in computational science, underpinning applications across a wide array of disciplines. Specifically, robust sampling facilitates the efficient exploration of high-dimensional phase spaces, which is essential for characterizing rare events [2, 3, 4], estimating

macroscopic thermodynamic observables [5, 6], and calculating free energy landscapes [7]. Furthermore, these sampling methodologies are deeply integral to advances in Bayesian data assimilation [8] and conformational sampling in structural biology [9].

Historically, approaches to this problem have focused on solving the stationary Fokker–Planck equation (SFPE) governing the distribution. Traditional deterministic techniques, such as finite difference or finite element methods [10, 11], inherently suffer from the *curse of dimensionality*. More recently, the success of deep neural networks has catalyzed the development of machine learning techniques to parameterize high-dimensional PDEs. Pioneering works like the *deep Ritz method* [12], *physics-informed neural networks* (PINNs) [13], and methods utilizing weak formulations such as *weak adversarial networks* (WAN) [14, 15, 16, 17], have shown promise in approximating the solution of the SFPE. However, these PDE-centric approaches primarily focus on approximating the density function itself. They do not inherently provide a direct, efficient mechanism for generating independent samples from the distribution, which remains a distinct and critical operational requirement in many practical scenarios.

To directly address the sampling requirement and circumvent the dimensionality limitations of grid-based methods, Monte Carlo (MC) methods have established themselves as the dominant paradigm. The standard approach involves numerically integrating the overdamped Langevin stochastic differential equation (SDE) [18, 19, 20]:

$$dX_t = -\nabla V(X_t)dt + \sqrt{2k_B\mathcal{T}}dB_t, \quad (2)$$

where B_t is a standard Brownian motion. Under the assumption of a strongly confining potential, the probability density of the trajectory X_t converges exponentially fast to the unique invariant Boltzmann distribution [21]. This simplicity and theoretical robustness have made Langevin dynamics the backbone of modern sampling algorithms [22, 23, 24].

However, the efficacy of standard Langevin dynamics is severely compromised when the potential landscape $V(x)$ is non-convex and exhibits multiple local minima. In such *metastable* regimes, particularly at low temperatures, the system remains trapped in local basins for exponentially long periods before crossing high energy barriers [25, 21]. Consequently, the simulation horizon required to achieve convergence becomes prohibitively large. While advanced techniques such as parallel tempering [26] or annealing [27] have been proposed to mitigate this, they often introduce significant algorithmic complexity or computational overhead.

Addressing this specific limitation, this work introduces a novel framework that fundamentally accelerates the exploration of complex landscapes. We propose a Lévy-score-based Monte Carlo method that leverages the heavy-tailed jumps characteristic of Lévy processes to directly bypass high energy barriers. Crucially, our approach retains the algorithmic simplicity and ease of implementation inherent to standard Monte Carlo methods, while drastically reducing the temporal cost required to escape metastable states.

Related works:

Fractional Langevin Monte Carlo. Fractional Langevin Monte Carlo was first introduced in [28, 29]. It has since demonstrated significant utility in modern machine learning, particularly in the domains of non-convex optimization and heavy-tailed sampling [30, 31, 32]. The fundamental principle underlying fractional Langevin Monte Carlo involves formulating a modified SDE of the form $dX_t = b(X_{t-})dt + dL_t^\alpha$. Specifically, the objective is to determine an appropriate drift coefficient b such that the resulting dynamics, driven by an α -stable Lévy process L_t^α , preserve the target invariant measure associated with (2). However, these foundational studies omit rigorous mathematical arguments demonstrating that the considered target measures are genuinely invariant under the proposed dynamics. Furthermore, they lack a formal proof regarding the uniqueness

of the stationary solution to Kolmogorov’s forward equation, relying instead on heuristic justifications rather than complete theoretical guarantees. Subsequent research has sought to generalize this framework. For instance, the authors of [33] extended the methodology to encompass general Lévy processes; however, their analytical scope was restricted to the specific setting of target distributions supported on the half-line and driven by compound Poisson noise. Additionally, recent studies [34, 35] have adapted the approach to investigate specific target invariant distributions exhibiting heavy-tailed properties.

Score-based generative models with α -stable Lévy processes. In [36], a class of score-based generative models was introduced, employing a tractable forward Lévy process defined by $X_t = a(t)X_0 + \gamma(t)\epsilon$. In this formulation, the initial state X_0 is drawn from the target data distribution, and the driving noise ϵ follows an isotropic α -stable distribution with a unit scale parameter. To rigorously construct the corresponding time-reversed Lévy dynamics, the authors define a fractional score function, $s_t^{(\alpha)}(x) = (\Delta^{\frac{\alpha-2}{2}} \nabla p_t(x))/p_t(x)$ for $1 < \alpha \leq 2$, associated with the marginal probability density $p_t(x)$. Computationally, this fractional score function is parameterized via a neural network. Its analytical structure is amenable to efficient approximation using denoising score matching, thereby closely paralleling the established training paradigms of standard score-based diffusion models.

Our Contributions

The primary objective of this work is to overcome the profound computational bottlenecks associated with sampling from metastable invariant distributions. Building upon our previous theoretical investigations, we develop a novel, mathematically rigorous Lévy-score-based Monte Carlo sampling framework. The main contributions of this paper are summarized as follows:

- **Robust and efficient method:** We propose a fundamentally new sampling dynamics by systematically injecting jump noise into the standard continuous Itô diffusion. Specifically, we construct a modified drift function that explicitly incorporates a Lévy score. This carefully engineered drift balances the introduced heavy-tailed jumps, ensuring that the augmented stochastic system still exactly preserves the target Boltzmann distribution as its invariant measure.
 1. **Accelerated transitions across metastable states:** We effectively resolve the critical challenge of slow mixing in non-convex potential landscapes. The explicitly introduced jump mechanisms allow the system’s trajectory to rapidly bypass high-energy barriers. This significantly accelerates transitions between isolated metastable states, fundamentally mitigating the exponentially long trapping times that plague classical continuous-path Langevin dynamics.
 2. **Algorithmic simplicity and empirical efficacy:** Despite the sophisticated theoretical machinery underlying the SDE construction, the resulting Monte Carlo scheme retains the exceptional algorithmic simplicity and ease of implementation characteristic of classical methods. Through extensive numerical experiments on benchmark problems exhibiting complex metastability, we empirically demonstrate the superior temporal efficiency, scalability, and robustness of our approach.
- **Rigorous theoretical guarantees:** A cornerstone of the proposed methodology lies in its rigorous mathematical foundation. For the newly constructed jump-augmented SDE, under some mild assumptions, we establish its global well-posedness and geometric ergodicity, alongside a detailed theoretical analysis of the mechanisms driving the accelerated sampling rate.

These theoretical guarantees are crucial, as they ensure that the time-averaged trajectory converges robustly to the target distribution, thereby providing a strict justification for the proposed Monte Carlo sampling algorithm (see Theorems 3.4, 3.7, and ??).

- **Numerical verification:** We conduct numerical experiments on both low and highdimensional problems, with the presence of meta-stable states. We compare our LSB-MC method with the traditional Itô diffusion-based Monte Carlo and demonstrate that the LSB-MC achieves comparable accuracy with a significantly lower computational cost. Furthermore, the LSB-MC can explore all the meta-stable states in both low temperature and high temperature scenarios

The remainder of this paper is organized as follows. Section 2 provides the preliminary mathematical background on Itô diffusions and the target Boltzmann distribution. In Section 3, we detail the construction of the jump-noise-augmented SDE, and the rigorous proofs of global well-posedness and ergodicity, and also the mechanism of acceleration is discussed. Section ?? presents comprehensive numerical experiments to validate the efficacy of the proposed Monte Carlo method. Finally, concluding remarks are given in Section ??.

2 Preliminaries

In this section, we introduce some preliminaries that will be used in our later analysis and discussion.

2.1 Reviews on diffusion-based Monte Carlo

We consider the following stochastic differential equation (SDE) on $\mathbb{R}^d$:

$$dX_t = -\nabla V(X_t)dt + \sigma dB_t, \quad (3)$$

where $V : \mathbb{R}^d \rightarrow \mathbb{R}$ is a potential function, σ is the constant noise intensity, and B_t is a w -dimensional standard Brownian motion. We commence by establishing the necessary assumptions regarding the potential V to ensure the well-posedness of the underlying dynamics.

Assumption 2.1. 1. *There exists a positive constant K_1 such that*

$$-2x \cdot \nabla V(x) \leq K_1(|x|^2 + 1) \quad \text{for all } x \in \mathbb{R}^d,$$

and V is locally uniformly continuous.

2. *There exists a positive constant δ_0 such that for all $R > 0$ and $x, z \in \mathbb{R}^d$ satisfying $|x| \leq R$, $|z| \leq R$, and $|x - z| \leq \delta_0$, the inequality*

$$2(x - z) \cdot (\nabla V(z) - \nabla V(x)) \leq K_R |x - z|^2$$

holds, where K_R is a positive constant dependent on R .

3. *There exist a positive constant α , a compact set $C \subset \mathbb{R}^d$, a measurable function $f : \mathbb{R}^d \mapsto [1, \infty)$, and a twice continuously differentiable function $U : \mathbb{R}^d \mapsto \mathbb{R}_+$ satisfying*

$$\mathcal{L}U(x) \leq -\alpha f(x) + 1_C(x) \quad \text{for all } x \in \mathbb{R}^d.$$

Under Assumption 2.1, the existence and uniqueness of the solution to the Itô SDE (3) are theoretically guaranteed. Furthermore, the process is ergodic and admits a unique invariant distribution p_∞ ; see, e.g., [37, Theorem 2.2, Theorem 2.4, Remark 5.5, Lemma 6.4]. The probability density function of the solution process X_t at time t and at position x , denoted by $p_t(x)$, evolves according to the Fokker–Planck equation

$$\partial_t p_t(x) = \nabla \cdot (\nabla V(x) p_t(x)) + \frac{\sigma^2}{2} \Delta p_t(x). \quad (4)$$

The invariant distribution p_∞ is defined as the long-time limit of the transient density p_t , namely, $p_\infty := \lim_{t \rightarrow +\infty} p_t$. The invariant measure of the SDE (3) admits the explicit analytical form

$$p_\infty(x) = Z^{-1} \exp \left\{ -\frac{2V(x)}{\sigma^2} \right\}, \quad (5)$$

which is widely recognized in statistical mechanics as the Boltzmann distribution. Here, Z is the normalization constant—commonly referred to as the partition function—ensuring that p_∞ integrates to unity.

The ergodicity of the solution process X_t implies that the invariant distribution p_∞ can be accurately approximated by the time average along a sufficiently long trajectory:

$$p_\infty(x) \approx \frac{1}{T} \int_0^T \delta(x - X_t) dt. \quad (6)$$

This ergodic property underpins Monte Carlo methods, such as the Euler–Maruyama scheme, facilitating the numerical sampling of the invariant distribution. However, in the weakly stochastic regime (i.e., when the noise intensity σ is small) and when the potential V features multiple deep wells separated by high energy barriers, the simulation time T required in (6) becomes prohibitively large. In such scenarios, the process X_t exhibits metastability, remaining trapped within local minima for exponentially long durations before transitioning to other wells—a phenomenon characteristic of rare events. This computational inefficiency arises from the continuous, diffusive nature of Brownian motion, which yields an average spatial displacement of order $\mathcal{O}(\sqrt{\Delta t})$ over a small time step Δt , thereby fundamentally limiting the exploration of the state space.

2.2 Generators and Dirichlet forms for diffusions and jump-diffusions

We collect the infinitesimal generators and the associated energy (Dirichlet) forms for (i) the overdamped Langevin diffusion and (ii) the jump–diffusions. Throughout, we work on the Hilbert space $L^2(p_\infty)$ with inner product

$$\langle f, g \rangle_{p_\infty} := \int_{\mathbb{R}^d} f(x) g(x) p_\infty(x) dx,$$

and use $C_c^\infty(\mathbb{R}^d)$ as a core of test functions.

1. Diffusion. The (backward) generator $\mathcal{L}$ of diffusion SDE (3) acts on $\varphi \in C_c^\infty(\mathbb{R}^d)$ as

$$(\mathcal{L}\varphi)(x) = -\nabla V(x) \cdot \nabla \varphi(x) + \frac{\sigma^2}{2} \Delta \varphi(x). \quad (7)$$

Definition 2.2 (Dirichlet form). *The associated Dirichlet form of diffusion SDE (3) is defined by*

$$\mathcal{E}_{\text{diff}}(f, g) := -\langle f, \mathcal{L}g \rangle_{p_\infty} = \frac{\sigma^2}{2} \int_{\mathbb{R}^d} \nabla f(x) \cdot \nabla g(x) p_\infty(x) dx, \quad f, g \in C_c^\infty(\mathbb{R}^d), \quad (8)$$

so that

$$\mathcal{E}_{\text{diff}}(f, f) = \frac{\sigma^2}{2} \int_{\mathbb{R}^d} |\nabla f(x)|^2 p_\infty(x) dx \geq 0.$$

Equivalently,

$$\mathcal{L}g = \frac{\sigma^2}{2} p_\infty^{-1} \nabla \cdot (p_\infty \nabla g), \quad (9)$$

which makes the $L^2(p_\infty)$ -symmetry of $\mathcal{E}_{\text{diff}}$ immediately.

2. Jump–diffusion. Consider the autonomous Lévy SDE

$$dZ_t = b(Z_{t-}) dt + \sigma dB_t + dL_t, \quad (10)$$

where L_t is a pure-jump Lévy process with Lévy measure $\nu(dr)$ [38]. We adopt the (forward) jump-operator

$$\int_{\mathbb{R}^d \setminus \{0\}} (p(x-r) - p(x)) \nu(dr),$$

namely the backward jump generator

$$(\mathcal{J}\varphi)(x) := \int_{\mathbb{R}^d \setminus \{0\}} (\varphi(x+r) - \varphi(x)) \nu(dr), \quad \varphi \in C_c^\infty(\mathbb{R}^d),$$

whenever the integral is well-defined. Hence the full generator $\mathcal{A}$ is

$$(\mathcal{A}\varphi)(x) = b(x) \cdot \nabla \varphi(x) + \frac{\sigma^2}{2} \Delta \varphi(x) + \int_{\mathbb{R}^d \setminus \{0\}} (\varphi(x+r) - \varphi(x)) \nu(dr). \quad (11)$$

Definition 2.3 (Dirichlet form). *Suppose p_∞ is invariant for (10), we define the Dirichlet form of jump-diffusion SDE (10) as*

$$\mathcal{E}_{\text{levy}}(f, g) := -\langle f, \mathcal{A}g \rangle_{p_\infty}, \quad f, g \in C_c^\infty(\mathbb{R}^d),$$

whenever the integrals are well-defined.

2.3 Mixing, spectral gap, and mixing-time comparison

In this section, we outline the theoretical framework for analyzing the *mixing* properties of the sampled processes—which characterizes the convergence of their probability distributions—through the lens of test functions and the spectral gap of the associated Markov semigroup. Specifically, we adopt an $L^2(p_\infty)$ -based criterion for mixing, which is equivalent to convergence in the χ^2 -divergence for probability densities, and connect it to the corresponding Poincaré constant. This formulation establishes a rigorous comparison principle between the standard diffusion sampler (3) and the proposed Lévy-score sampler (10). Provided both processes admit the identical invariant measure p_∞ , comparing their convergence rates mathematically reduces to evaluating their respective $L^2(p_\infty)$ spectral gaps.

Definition 2.4 (Semigroups, and invariance). *Let $(P_t^{\text{diff}})_{t \geq 0}$ and $(P_t^{\text{levy}})_{t \geq 0}$ be the Markov semigroups induced by (3) and (10), respectively:*

$$(P_t^{\text{diff}} f)(x) := \mathbb{E}[f(X_t) \mid X_0 = x], \quad (P_t^{\text{levy}} f)(x) := \mathbb{E}[f(Z_t) \mid Z_0 = x].$$

We assume that p_∞ is invariant for both dynamics (3) and (10); in particular,

$$p_\infty(P_t f) = p_\infty(f), \quad t \geq 0, \quad (12)$$

for all bounded measurable f for which the expectations are well-defined.

Throughout, we work on $L^2(p_\infty)$ with inner product

$$\langle f, g \rangle_{p_\infty} := \int_{\mathbb{R}^d} f(x) g(x) p_\infty(x) dx, \quad \|f\|_{L^2(p_\infty)}^2 := \langle f, f \rangle_{p_\infty},$$

and define

$$p_\infty(f) := \int_{\mathbb{R}^d} f(x) p_\infty(x) dx, \quad \text{Var}_{p_\infty}(f) := \|f - p_\infty(f)\|_{L^2(p_\infty)}^2.$$

From distributional distance to test functions (why “test functions” appear). A difference of probability measures $\mu - \nu$ is a linear functional acting on test functions:

$$(\mu - \nu)(f) = \int f d\mu - \int f d\nu.$$

Many common metrics can be written as a supremum over a class of test functions; for example, $\|\mu - \nu\|_{\text{TV}} = \sup_{\|f\|_\infty \leq 1} |(\mu - \nu)(f)|$ and $W_1(\mu, \nu) = \sup_{\text{Lip}(f) \leq 1} |(\mu - \nu)(f)|$. In this paper we adopt an $L^2(p_\infty)$ -based notion: if $\mu \ll p_\infty$ and we write $h := \frac{d\mu}{dp_\infty}$, then the χ^2 -divergence is

$$\chi^2(\mu \| p_\infty) = \int (h - 1)^2 dp_\infty = \|h - 1\|_{L^2(p_\infty)}^2.$$

Thus controlling $\|h_t - 1\|_{L^2(p_\infty)}$ is a concrete way to quantify convergence of the law at time t to p_∞ , provided the evolving law stays absolutely continuous w.r.t. p_∞ . The spectral-gap method is precisely an L^2 contraction argument for a semigroup, and therefore naturally expressed in terms of test functions.

Definition 2.5 (L^2 energy identity). *Let $(P_t)_{t \geq 0}$ denote either (P_t^{diff}) or (P_t^{levy}) , with generator $\mathcal{G} \in \{\mathcal{L}, \mathcal{A}\}$ acting on the core $C_c^\infty(\mathbb{R}^d)$ introduced in Section 2.2. For $f \in C_c^\infty(\mathbb{R}^d)$ define the centered function*

$$g_t := P_t f - p_\infty(f).$$

Proposition 2.6. *The centered function $g_t \in L^2(p_\infty)$ and it can be shown that*

$$\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 = 2 \text{Re} \langle g_t, \mathcal{G} g_t \rangle_{p_\infty} = -2 \text{Re} \langle g_t, -\mathcal{G} g_t \rangle_{p_\infty}. \quad (13)$$

Proof. For any $f \in C_c^\infty(\mathbb{R}^d)$ and set $g_t := P_t f - p_\infty(f)$. Since $p_\infty(f)$ is constant in t and P_t satisfies the Kolmogorov backward equation on the core, we have $\partial_t g_t = \partial_t (P_t f) = \mathcal{G} P_t f$. Moreover, $\mathcal{G}$ annihilates constants, hence $\mathcal{G} g_t = \mathcal{G} P_t f$, and therefore $\partial_t g_t = \mathcal{G} g_t$. Differentiating the squared norm gives

$$\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 = \frac{d}{dt} \langle g_t, g_t \rangle_{p_\infty} = \langle \partial_t g_t, g_t \rangle_{p_\infty} + \langle g_t, \partial_t g_t \rangle_{p_\infty} = \langle \mathcal{G} g_t, g_t \rangle_{p_\infty} + \langle g_t, \mathcal{G} g_t \rangle_{p_\infty}.$$

Taking real parts yields $\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 = 2 \text{Re} \langle g_t, \mathcal{G} g_t \rangle_{p_\infty}$, as claimed. $\square$

Definition 2.7 (Dissipative quadratic form (coercivity)). *For a (possibly non-reversible) Markov semigroup with generator $\mathcal{G}$ on the core $C_c^\infty(\mathbb{R}^d)$, we define the dissipative quadratic form*

$$\mathcal{E}_{\mathcal{G}}(f, f) := -\operatorname{Re} \langle f, \mathcal{G}f \rangle_{p_\infty}, \quad f \in C_c^\infty(\mathbb{R}^d). \quad (14)$$

When $\mathcal{G}$ is $L^2(p_\infty)$ -reversible, $\mathcal{E}_{\mathcal{G}}(f, f) = -\langle f, \mathcal{G}f \rangle_{p_\infty}$ is the usual (symmetric) Dirichlet form.

Definition 2.8 ($L^2(p_\infty)$ coercivity (Poincaré) constant). *We define*

$$\lambda(\mathcal{G}) := \inf \left\{ \frac{\mathcal{E}_{\mathcal{G}}(h, h)}{\|h\|_{L^2(p_\infty)}^2} : h \in C_c^\infty(\mathbb{R}^d), p_\infty(h) = 0, h \neq 0 \right\}. \quad (15)$$

Equivalently, $\lambda(\mathcal{G}) > 0$ is the largest constant such that

$$\mathcal{E}_{\mathcal{G}}(h, h) \geq \lambda(\mathcal{G}) \|h\|_{L^2(p_\infty)}^2, \quad h \in C_c^\infty(\mathbb{R}^d), p_\infty(h) = 0. \quad (16)$$

In the reversible case, $\lambda(\mathcal{G})$ coincides with the standard L^2 spectral gap.

Definition 2.9 (Poincaré inequality). *By Definition 2.3, $\lambda(\mathcal{G}) > 0$ is the largest constant such that*

$$\mathcal{E}(h, h) \geq \lambda(\mathcal{G}) \operatorname{Var}_{p_\infty}(h), \quad h \in C_c^\infty(\mathbb{R}^d), \quad (17)$$

holds.

Proposition 2.10 (Exponential L^2 convergence from a spectral gap). *Assume that (17) holds with $\lambda(\mathcal{G}) > 0$. Then for all $f \in C_c^\infty(\mathbb{R}^d)$,*

$$\|P_t f - p_\infty(f)\|_{L^2(p_\infty)} \leq e^{-\lambda(\mathcal{G})t} \|f - p_\infty(f)\|_{L^2(p_\infty)}, \quad t \geq 0. \quad (18)$$

Proof. Let $g_t = P_t f - p_\infty(f)$ as above. From (13) and (??),

$$\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 = -2 \operatorname{Re} \mathcal{E}(g_t, g_t).$$

Since $\mathcal{E}(h, h) \geq 0$ for all h and $\mathcal{E}(h, h) \in \mathbb{R}$ for the Dirichlet forms considered here, this is

$$\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 = -2 \mathcal{E}(g_t, g_t).$$

Applying the Poincaré inequality (17) to $h = g_t$ yields

$$\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 \leq -2 \lambda(\mathcal{G}) \operatorname{Var}_{p_\infty}(g_t).$$

By invariance (12), $p_\infty(g_t) = p_\infty(P_t f) - p_\infty(f) = 0$, hence $\operatorname{Var}_{p_\infty}(g_t) = \|g_t\|_{L^2(p_\infty)}^2$. Therefore

$$\frac{d}{dt} \|g_t\|_{L^2(p_\infty)}^2 \leq -2 \lambda(\mathcal{G}) \|g_t\|_{L^2(p_\infty)}^2.$$

Gronwall's inequality implies $\|g_t\|_{L^2(p_\infty)}^2 \leq e^{-2\lambda(\mathcal{G})t} \|g_0\|_{L^2(p_\infty)}^2$, which is equivalent to (18). $\square$

Definition 2.11 (L^2 mixing time). *For any $\delta \in (0, 1)$, define the $L^2(p_\infty)$ mixing time of (P_t) by*

$$t_{\text{mix}}^{(2)}(\delta) := \inf \left\{ t \geq 0 : \|P_t f - p_\infty(f)\|_{L^2(p_\infty)} \leq \delta \|f - p_\infty(f)\|_{L^2(p_\infty)} \quad \forall f \in L^2(p_\infty) \right\}. \quad (19)$$

The estimate (18) yields the standard bound

$$t_{\text{mix}}^{(2)}(\delta) \leq \frac{1}{\lambda(\mathcal{G})} \log \frac{1}{\delta}. \quad (20)$$

Now we are ready for convenient comparison principles between the diffusion sampler (3) and the Lévy-score sampler (10) (when they share the same invariant measure p_∞).

1. **(Connection to χ^2 -mixing of distributions.)** If an initial law μ is absolutely continuous with respect to p_∞ , write $h = \frac{d\mu}{dp_\infty} \in L^2(p_\infty)$, and let $\mu_t := \mu P_t$ denote the law at time t . Under mild regularity assumptions ensuring that $\mu_t \ll p_\infty$ and that the adjoint semigroup P_t^* acts on $L^2(p_\infty)$, one has

$$\frac{d\mu_t}{dp_\infty} = P_t^* h =: h_t.$$

Since $\chi^2(\mu_t \| p_\infty) = \|h_t - 1\|_{L^2(p_\infty)}^2$, this implies

$$\chi^2(\mu_t \| p_\infty) = \|h_t - 1\|_{L^2(p_\infty)}^2 = \|P_t^*(h - 1)\|_{L^2(p_\infty)}^2.$$

Note that $h - 1$ has mean zero with respect to p_∞ . The $L^2(p_\infty)$ spectral gap bound (18) provides exponential contraction of P_t on the mean-zero subspace of $L^2(p_\infty)$; moreover, P_t and its adjoint P_t^* have the same $L^2(p_\infty)$ operator norm, so the same exponential contraction estimate holds for P_t^* . Applying this to the mean-zero function $h - 1$ yields

$$\chi^2(\mu_t \| p_\infty)^{1/2} = \|P_t^*(h - 1)\|_{L^2(p_\infty)} \leq e^{-\lambda(\mathcal{G})t} \|h - 1\|_{L^2(p_\infty)} = e^{-\lambda(\mathcal{G})t} \chi^2(\mu \| p_\infty)^{1/2}.$$

Thus the L^2 -mixing time $t_{\text{mix}}^{(2)}$ corresponds to exponential decay of $\chi^2(\mu_t \| p_\infty)$ for densities.

2. **(Mixing-time comparison principle.)** At the level of the $L^2(p_\infty)$ contraction bound, comparing the convergence speed of the two samplers reduces to comparing their spectral gaps:

$$\lambda(\mathcal{A}) \quad \text{versus} \quad \lambda(\mathcal{L}).$$

In particular, if $\lambda(\mathcal{A}) \geq c \lambda(\mathcal{L})$ for some $c > 1$, then (20) implies

$$t_{\text{mix, Levy}}^{(2)}(\delta) \leq c^{-1} t_{\text{mix, diff}}^{(2)}(\delta).$$

Remark 2.12. The $L^2(p_\infty)$ approach controls a χ^2 -type distance and is often convenient for spectral comparisons. Other distances (e.g. total variation or Wasserstein) require different tools (e.g. minorization/coupling or contractive couplings), and a spectral gap alone does not automatically give sharp bounds in those metrics without additional assumptions.

3 Methodology

3.1 Heuristics, key ideas and jump-augmented SDE

In contrast to Brownian motion, non-Gaussian Lévy processes (e.g., α -stable Lévy processes) are characterized by discontinuous trajectories with non-local jump displacements. These macroscopic spatial excursions significantly enhance transition probabilities across high energy barriers, effectively mitigating the metastability issue. Leveraging the Lévy score function framework [39, 40], it is feasible to incorporate non-local jump processes into the underlying SDE (3) while strictly

preserving the target invariant Boltzmann distribution p_∞ . Specifically, by employing the Lévy score function, the evolution of the probability density p_t can be reformulated via the following integro-differential Lévy–Fokker–Planck equation (LFPE):

$$\begin{aligned}\partial_t p_t(x) &= \nabla \cdot (\nabla V p_t)(x) + \frac{\sigma^2}{2} \Delta p_t(x) \\ &= \nabla \cdot \left[\left(\nabla V(x) + \int_0^1 \int_{\mathbb{R}^d - \{0\}} \frac{r p_t(x - \theta r)}{p_t(x)} \nu(dr) d\theta \right) p_t(x) \right] + \frac{\sigma^2}{2} \Delta p_t(x) \\ &\quad + \int_{\mathbb{R}^d - \{0\}} (p_t(x - r) - p_t(x)) \nu(dr),\end{aligned}\tag{21}$$

where ν represents an appropriate Lévy measure. Here, the expression

$$S_L(t, x) := - \int_0^1 \int_{\mathbb{R}^d - \{0\}} \frac{r p_t(x - \theta r)}{p_t(x)} \nu(dr) d\theta,\tag{22}$$

defines the Lévy score function [40], a pivotal concept in the analysis of probability flow and entropy production for jump-diffusion processes [41]. Treating the Lévy score function $S_L(t, x)$ as an a priori known term, the LFPE structure implies that the probability flow p_t corresponds to the following associated SDE (cf. [39, Theorem 2.3]):

$$dY_t = \left(-\nabla V(Y_{t-}) - \int_0^1 \int_{\mathbb{R}^d - \{0\}} \frac{r p_t(Y_{t-} - \theta r)}{p_t(Y_{t-})} \nu(dr) d\theta \right) dt + \sigma dB_t + dL_t, \quad Y_0 \sim p_0,\tag{23}$$

where L_t is a Lévy process independent of both the Brownian motion B_t and the initial distribution p_0 , driven by a Lévy measure ν governed by the score function S_L in (22).

Although the diffusion process X_t in (3) and the jump-diffusion process Y_t in (23) exhibit distinct pathwise behaviors, they inherently share the identical probability flow. Consequently, utilizing the time-dependent jump-diffusion SDE (23) to sample the invariant measure offers no theoretical acceleration compared to the diffusion SDE (3). This limitation stems from the density-dependent Lévy score function $S_L(t, x)$ effectively counteracting the macroscopic jumps of the Lévy process L_t within the probability space. However, as the system relaxes toward its stationary state, the Lévy score converges to a time-independent asymptotic form:

$$\begin{aligned}S_L^s(x) &:= - \int_0^1 \int_{\mathbb{R}^d - \{0\}} \frac{r p_\infty(x - \theta r)}{p_\infty(x)} \nu(dr) d\theta \\ &= - \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \exp \left\{ -\frac{2V(x - \theta r)}{\sigma^2} + \frac{2V(x)}{\sigma^2} \right\} \nu(dr) d\theta.\end{aligned}\tag{24}$$

This observation motivates the proposal of the following autonomous jump-diffusion SDE for accelerated sampling purposes:

$$(\text{jump-argmented SDE}) \quad dZ_t = (-\nabla V(Z_{t-}) + S_L^s(Z_{t-})) dt + \sigma dB_t + dL_t.\tag{25}$$

Prior to discussing the numerical sampling strategy, we establish the theoretical properties of the proposed dynamics (25). To facilitate the analysis, we require additional structural properties for the potential function V , as detailed in the following assumption.

Assumption 3.1. *The potential V is coercive, satisfying $V(x) \rightarrow \infty$ as $|x| \rightarrow \infty$. Furthermore, we impose a monotonicity condition on the gradient magnitude: there exists a constant $R > 0$ such that*

$$|\nabla V(x_1)| \geq |\nabla V(x_2)| \quad \text{whenever} \quad |x_1| \geq |x_2| > R.$$

Remark 3.2. *Assumption 3.1 is analogous to conditions employed in the study of Γ -convergence [42]. In the present context, this assumption is crucial for analyzing the well-posedness of the jump-diffusion SDE (25), as it guarantees that the potential V possesses sufficient geometric regularity and growth properties at infinity.*

To address technical considerations regarding potentially unbounded drifts, we analyze the jump-diffusion SDE (25) through a sequence of truncated approximations. For any fixed $M > 0$, let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a smooth scalar cutoff function defined by

$$h(s) := \begin{cases} 1, & s \leq M - 1, \\ 0, & s \geq M. \end{cases} \quad (26)$$

We introduce an auxiliary truncated potential V^M associated with V as follows:

$$V^M(x) := h(U(x))U(x) + [1 - h(U(x))] |x|^2, \quad x \in \mathbb{R}^d. \quad (27)$$

Consider the corresponding SDE governed by this truncated potential:

$$dX_t^M = -\nabla V^M(X_t^M)dt + \sigma dB_t. \quad (28)$$

It is straightforward to verify that the modified drift term satisfies the coercivity and monotonicity properties outlined in Assumption 2.1. The associated invariant Boltzmann distribution for this truncated system reads

$$p_\infty^M(x) = (Z^M)^{-1} \exp \left\{ -\frac{2V^M(x)}{\sigma^2} \right\}. \quad (29)$$

Consequently, the global well-posedness and the existence of a unique invariant measure for the SDE (28) are guaranteed. Analogous to the derivation in the preceding section, we define the truncated jump-diffusion SDE corresponding to (25), which recovers the original dynamics in the limit as $M \rightarrow \infty$:

$$dZ_t^M = (-\nabla V^M(Z_{t-}^M) + S_L^{sM}(Z_{t-}^M)) dt + \sigma dB_t + dL_t, \quad (30)$$

where

$$S_L^{sM}(x) := - \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \exp \left\{ -\frac{2V^M(x - \theta r)}{\sigma^2} + \frac{2V^M(x)}{\sigma^2} \right\} \nu(dr) d\theta. \quad (31)$$

3.2 Global well-posedness and ergodicity of jump-argmented SDE (25)

We proceed under the following additional structural assumptions regarding the Lévy measure ν .

Assumption 3.3. *The Lévy measure ν possesses a finite second moment, i.e., $\int_{\mathbb{R}^d - \{0\}} |r|^2 \nu(dr) < \infty$.*

Theorem 3.4. *Under Assumptions 2.1, 3.1, and 3.3, the jump-diffusion SDE (25) is globally well-posed.*

Proof. Observe that for $x \in \mathbb{R}^d$ satisfying $V(x) > M$, the drift term involving the modified Lévy score $S_L^{sM}(x)$ satisfies the following estimate:

$$\begin{aligned}
& -2 \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \cdot x \exp \left\{ -\frac{2V^M(x - \theta r)}{\sigma^2} + \frac{2V^M(x)}{\sigma^2} \right\} \nu(dr) d\theta \\
&= -2 \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \cdot x \exp \left\{ -\frac{2|x - \theta r|^2}{\sigma^2} + \frac{2|x|^2}{\sigma^2} \right\} \nu(dr) d\theta \\
&= -2 \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \cdot x \exp \left\{ \frac{4x \cdot \theta r - 2|\theta r|^2}{\sigma^2} \right\} \nu(dr) d\theta \\
&\leq 8|x|^2 \int_0^1 \int_{\mathbb{R}^d - \{0\}} \frac{|\theta r|^2}{\sigma^2} \exp \left\{ -\frac{2|\theta r|^2}{\sigma^2} \right\} \nu(dr) d\theta.
\end{aligned} \tag{32}$$

Because the potential V is coercive (Assumption 3.1), the sublevel set $\{x \in \mathbb{R}^d \mid V(x) \leq M\}$ is compact. Consequently, the continuous function $x \cdot S_L^{sM}(x)$ is bounded on this set, yielding

$$\sup_{\{x \in \mathbb{R}^d \mid V(x) \leq M\}} |x \cdot S_L^{sM}(x)| < \infty. \tag{33}$$

Let us define the constant K'_1 as

$$K'_1 := K_1 + \sup_{\{x \in \mathbb{R}^d \mid V(x) \leq M\}} |x \cdot S_L^{sM}(x)| + 8 \int_0^1 \int_{\mathbb{R}^d - \{0\}} \frac{|\theta r|^2}{\sigma^2} \exp \left\{ -\frac{2|\theta r|^2}{\sigma^2} \right\} \nu(dr) d\theta < \infty, \tag{34}$$

where K_1 is the constant introduced in Assumption 2.1. Then, for all $x \in \mathbb{R}^d$, we obtain the global bound

$$2x \cdot (-\nabla V^M(x) + S_L^{sM}(x)) \leq K'_1(|x|^2 + 1). \tag{35}$$

Thus, by invoking [37, Theorem 2.2], we conclude that the solution process of the truncated system (30) is non-explosive. Furthermore, it is straightforward to verify that the modified drift $-\nabla V^M + S_L^{sM}$ is locally Lipschitz continuous. The local Lipschitz continuity of the drift term guarantees the uniqueness of the solution to (30). Taking the limit as $M \rightarrow \infty$, this sequence of well-posed local approximations systematically extends to a globally well-posed solution for the original jump-diffusion SDE (25), which completes the proof. $\square$

To further guarantee that the jump-diffusion SDE (25) possesses well-defined transition densities and a unique invariant distribution, we introduce the following regularity assumption.

Assumption 3.5.

1. The potential V is infinitely differentiable, i.e., $V \in C^\infty(\mathbb{R}^d, \mathbb{R})$.
2. There exists a constant $\delta > 0$ such that

$$\limsup_{\epsilon \rightarrow 0} \frac{\left(\int_{|r| \leq \epsilon} |r|^3 \nu(dr) \right)^{1/3}}{\left(\int_{|r| \leq \epsilon} |r|^2 \nu(dr) \right)^{(1+\delta)/2}} < \infty.$$

Remark 3.6. Assumption 3.5 is introduced to mathematically ensure the existence of smooth transition densities for the jump-diffusion SDE (25). Alternative, simpler conditions can also be employed to achieve this. For instance, consider the scenario where the Lévy process is a compound Poisson process governed by the Lévy measure $\lambda_0 \nu_J$, where λ_0 acts as the intensity parameter and ν_J characterizes the distribution of the jump size. In this case, it is sufficient for ν_J to possess bounded support and a smooth density $\nu_J \in C^\infty(\mathbb{R}^d, \mathbb{R})$; for further details and proofs, we refer the reader to [43, Theorem 2-29].

Theorem 3.7. Under Assumptions 2.1, 3.1, 3.3, and 3.5, the jump-diffusion SDE (25) admits a unique invariant measure p_∞ , which precisely corresponds to the Boltzmann distribution of the SDE (3), in the sense that for every compact set $K \subset \mathbb{R}^d$,

$$\lim_{M \rightarrow \infty} \sup_{x \in K} |p_\infty^M(x) - p_\infty(x)| = 0. \quad (36)$$

Proof. By construction, the truncated auxiliary potential V^M defined in (27) is infinitely differentiable. Furthermore, although the drift vector field $-\nabla V^M$ is not necessarily globally bounded, all of its higher-order derivatives remain bounded. Consequently, under Assumptions 2.1, 3.3, and 3.5, the solution process Z_t^M to the truncated jump-diffusion SDE (30) admits unique C^∞ transition probability densities, a regularity property rigorously established in [44, Theorem 3.4]. This inherent regularity ensures that the stationary distribution p_∞^M defined in (29) acts as the unique invariant measure for the truncated process Z_t^M . Specifically, this holds because p_∞^M solves the stationary LFPE associated with (30):

$$\begin{aligned} 0 &= \nabla \cdot \left[\left(\nabla V^M(x) + \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \exp \left\{ -\frac{2V^M(x - \theta r)}{\sigma^2} + \frac{2V^M(x)}{\sigma^2} \right\} \nu(dr) d\theta \right) p_\infty^M(x) \right] \\ &\quad + \frac{\sigma^2}{2} \Delta p_\infty^M(x) + \int_{\mathbb{R}^d - \{0\}} (p_\infty^M(x - r) - p_\infty^M(x)) \nu(dr) \\ &= \nabla \cdot \left[\left(\nabla V^M(x) + \frac{\sigma^2}{2} \nabla \log p_\infty^M(x) \right) p_\infty^M(x) \right]. \end{aligned} \quad (37)$$

Taking the limit as $M \rightarrow \infty$ yields the desired uniform convergence on compact sets to the target distribution p_∞ , thereby completing the proof. $\square$

3.3 Spectral analysis for jump-augmented SDE (25)

In this section, we proceed to analyze the underlying theoretical mechanisms driving the accelerated sampling rate of the proposed methodology.

3.3.1 Closed-form Dirichlet form of jump-augmented SDE (25)

The following assumption is imposed *only* to justify the explicit energy identity below under the non-truncated convention (11).

Assumption 3.8 (Finite activity). *The Lévy measure satisfies $\nu(\mathbb{R}^d \setminus \{0\}) < \infty$.*

Remark 3.9 (On truncation conventions). *Assumption 3.8 (compound Poisson/finite activity) together with Assumption 3.3 ensures that the non-truncated jump generator in (11) is well-defined on C_c^∞ and justifies repeated use of Tonelli/Fubini in the proof below. If one instead works with the standard Lévy–Itô truncation $\int (\varphi(x + r) - \varphi(x) - \nabla \varphi(x) \cdot r \mathbf{1}_{\{|r| \leq 1\}}) \nu(dr)$, then the forward equation and the energy identity acquire the corresponding compensator terms.*

Under Assumptions 3.3 and 3.8, it can be shown that the Lévy-score drift of jump-diffusion SDE (25) cancels the non-symmetric part induced by the pure-jump generator at the level of the quadratic form. Consequently, the energy admits a closed-form decomposition into a diffusion part and a jump increment part.

Proposition 3.10 (Closed-form Dirichlet energy). *Assume Assumptions 3.3 and 3.8. Let $\mathcal{A}$ be given by (11) with $b(x) = -\nabla V(x) + S_L^s(x)$, where S_L^s is defined by (24) and $\mathcal{E}_{\text{levy}}(f, g) := -\langle f, \mathcal{A}g \rangle_{p_\infty}$. Then for any $f \in C_c^\infty(\mathbb{R}^d)$,*

$$\mathcal{E}_{\text{levy}}(f, f) = \frac{\sigma^2}{2} \int_{\mathbb{R}^d} |\nabla f(x)|^2 p_\infty(x) dx + \frac{1}{2} \iint_{\mathbb{R}^d \times (\mathbb{R}^d \setminus \{0\})} (f(x+r) - f(x))^2 p_\infty(x) \nu(dr) dx. \quad (38)$$

In particular, $\mathcal{E}_{\text{levy}}(f, f) \geq 0$.

Remark 3.11. For any $f \in C_c^\infty(\mathbb{R}^d)$, the quadratic form in Proposition 3.10 satisfies $\mathcal{E}_\mathcal{A}(f, f) = -\text{Re}\langle f, \mathcal{A}f \rangle_{p_\infty} = \mathcal{E}_{\text{levy}}(f, f) \geq 0$.

Proof. For any $f \in C_c^\infty(\mathbb{R}^d)$, by definition, 2.2 we know that

$$\begin{aligned} \mathcal{E}_{\text{levy}}(f, f) &= - \int_{\mathbb{R}^d} f(x) b(x) \cdot \nabla f(x) p_\infty(x) dx - \frac{\sigma^2}{2} \int_{\mathbb{R}^d} f(x) \Delta f(x) p_\infty(x) dx \\ &\quad - \int_{\mathbb{R}^d} f(x) (\mathcal{J}f)(x) p_\infty(x) dx. \end{aligned}$$

Step 1: diffusion contribution. Using $p_\infty(x) \propto \exp\{-2V(x)/\sigma^2\}$ and integration by parts (valid since f is compactly supported), the potential drift $-\nabla V$ cancels with the Δ -term to yield

$$- \int f(-\nabla V) \cdot \nabla f p_\infty dx - \frac{\sigma^2}{2} \int f \Delta f p_\infty dx = \frac{\sigma^2}{2} \int |\nabla f|^2 p_\infty dx.$$

Step 2: jump term and score cancellation. By expansion of $\mathcal{J}$,

$$- \int f(x) (\mathcal{J}f)(x) p_\infty(x) dx = \iint (f(x)^2 - f(x)f(x+r)) p_\infty(x) \nu(dr) dx. \quad (39)$$

Next, by the definition of S_L^s (24) and Tonelli's theorem (Assumptions 3.8 and 3.3),

$$- \int f(x) S_L^s(x) \cdot \nabla f(x) p_\infty(x) dx = \int_{\mathbb{R}^d} \int_0^1 \int_{\mathbb{R}^d \setminus \{0\}} f(x) r \cdot \nabla f(x) p_\infty(x - \theta r) \nu(dr) d\theta dx.$$

For fixed $r \neq 0$ and $\theta \in [0, 1]$, set $y = x - \theta r$ to obtain

$$\int_{\mathbb{R}^d} f(x) r \cdot \nabla f(x) p_\infty(x - \theta r) dx = \int_{\mathbb{R}^d} f(y + \theta r) r \cdot \nabla f(y + \theta r) p_\infty(y) dy.$$

Integrating over $\theta \in [0, 1]$ and using

$$\frac{d}{d\theta} \frac{1}{2} f(y + \theta r)^2 = f(y + \theta r) r \cdot \nabla f(y + \theta r),$$

we obtain

$$\int_0^1 f(y + \theta r) r \cdot \nabla f(y + \theta r) d\theta = \frac{1}{2} (f(y+r)^2 - f(y)^2).$$

Therefore,

$$-\int f S_L^s \cdot \nabla f p_\infty dx = \frac{1}{2} \iint (f(x+r)^2 - f(x)^2) p_\infty(x) \nu(dr) dx. \quad (40)$$

Summing (39) and (40) gives

$$-\int f S_L^s \cdot \nabla f p_\infty dx - \int f (\mathcal{J}f) p_\infty dx = \frac{1}{2} \iint (f(x+r) - f(x))^2 p_\infty(x) \nu(dr) dx.$$

Combining this with Step 1 yields (38). $\square$

For completeness, we define the symmetric part of the full bilinear form as follows.

Definition 3.12 (Symmetric bilinear form). *The symmetric part of the full bilinear form is given by*

$$\mathcal{E}_{\text{levy}}^{\text{sym}}(f, g) := \frac{1}{2} \left(\mathcal{E}_{\text{levy}}(f, g) + \mathcal{E}_{\text{levy}}(g, f) \right), \quad f, g \in C_c^\infty(\mathbb{R}^d).$$

Then Proposition 3.10 implies

$$\mathcal{E}_{\text{levy}}^{\text{sym}}(f, f) = \mathcal{E}_{\text{levy}}(f, f) = \frac{\sigma^2}{2} \int_{\mathbb{R}^d} |\nabla f|^2 p_\infty dx + \frac{1}{2} \iint (f(x+r) - f(x))^2 p_\infty(x) \nu(dr) dx.$$

3.3.2 $\lambda(\mathcal{A}) \geq \lambda(\mathcal{L})$

$$\mathcal{E}_{\text{levy}}(f, f) = \mathcal{E}_{\text{diff}}(f, f) + \mathcal{E}_{\text{jump}}(f, f),$$

where

$$\mathcal{E}_{\text{jump}}(f, f) := \frac{1}{2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d \setminus \{0\}} (f(x+r) - f(x))^2 p_\infty(x) \nu(dr) dx \geq 0.$$

for any $h \neq 0$, we have

$$\frac{\mathcal{E}_{\text{levy}}(h, h)}{\|h\|^2} \geq \frac{\mathcal{E}_{\text{diff}}(h, h)}{\|h\|^2}.$$

So

$$\lambda(\mathcal{A}) \geq \lambda(\mathcal{L}).$$

3.4 2-well

3.4.1 Partition of State Space

Basins of attraction and geometric partition. Assume that the energy landscape $V : \mathbb{R}^d \rightarrow \mathbb{R}$ is of double-well type and admits exactly two distinct local minima (well bottoms) $m_1, m_2 \in \mathbb{R}^d$. Consider the deterministic gradient flow

$$\dot{x}_t = -\nabla V(x_t), \quad x_0 = x \in \mathbb{R}^d,$$

and denote by $x_t(x)$ the (maximal) solution starting from x . The (forward) basin of attraction of m_i is defined by

$$\tilde{\Omega}_i := \left\{ x \in \mathbb{R}^d : x_t(x) \text{ exists for all } t \geq 0 \text{ and } \lim_{t \rightarrow \infty} x_t(x) = m_i \right\}, \quad i \in \{1, 2\}.$$

We assume that there are no other recurrent attractors so that, up to a negligible set of initial conditions, each point flows to one of the two minima. More precisely, there exists a Borel set $\mathcal{N} \subset \mathbb{R}^d$ with

$$\mathcal{N} \subset \partial \tilde{\Omega}_1 \cap \partial \tilde{\Omega}_2, \quad \mathcal{N} \text{ has Lebesgue and } p_\infty\text{-measure zero,}$$

such that

$$\mathbb{R}^d = \tilde{\Omega}_1 \cup \tilde{\Omega}_2 \cup \mathcal{N}, \quad \tilde{\Omega}_1 \cap \tilde{\Omega}_2 = \emptyset.$$

For definiteness we remove the null set and work with the measurable partition

$$\Omega_i := \tilde{\Omega}_i \setminus \mathcal{N}, \quad i \in \{1, 2\},$$

so that

$$\Omega_1 \cup \Omega_2 = \mathbb{R}^d, \quad \Omega_1 \cap \Omega_2 = \emptyset.$$

Invariant measure and well-weights. Let p_∞ denote the invariant density on $\mathbb{R}^d$ and define the well-weights

$$w_i := \int_{\Omega_i} p_\infty(x) dx, \quad i \in \{1, 2\}.$$

Clearly,

$$w_1 + w_2 = 1.$$

Conditional invariant measures on each region. Define the conditional densities (and the associated probability measures) by

$$p_\infty^{(i)}(x) := \frac{1}{w_i} p_\infty(x) \mathbf{1}_{\Omega_i}(x), \quad p_\infty^{(i)}(dx) := p_\infty^{(i)}(x) dx.$$

For any integrable test function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, define the conditional means

$$\bar{f}_{\Omega_i} := \int_{\Omega_i} f(x) p_\infty^{(i)}(dx).$$

Define the conditional variances

$$\text{Var}_{p_\infty^{(i)}}(f) := \int_{\Omega_i} (f(x) - \bar{f}_{\Omega_i})^2 p_\infty^{(i)}(dx).$$

Exact variance decomposition (two-state coarse graining). With the above partition, the global variance admits the exact decomposition

$$\text{Var}_{p_\infty}(f) = \sum_{i=1}^2 w_i \text{Var}_{p_\infty^{(i)}}(f) + w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2. \quad (41)$$

This is the (exact) “within-well + between-well” decomposition: the first term measures fluctuations inside each region Ω_i , while the second term measures the mismatch of the two regional averages.

3.4.2 Local Cores Inside Each Well

Local minima and core sets. Let $m_1, m_2 \in \mathbb{R}^d$ denote the two local minima (well bottoms) and define their distance

$$D := |m_2 - m_1|.$$

Fix a core radius $r_c > 0$ and define the core sets

$$\mathcal{C}_i := B(m_i, r_c), \quad i \in \{1, 2\}.$$

Assume the cores lie in their respective regions:

$$\mathcal{C}_1 \subset \Omega_1, \quad \mathcal{C}_2 \subset \Omega_2,$$

and (for convenience in the shell argument) assume also $D > 4r_c$ so that $\mathcal{C}_1 \cap \mathcal{C}_2 = \emptyset$.

Core masses and core-conditional measures. Define the core masses

$$c_i := \int_{\mathcal{C}_i} p_\infty(x) dx, \quad i \in \{1, 2\},$$

and define the core-conditional densities

$$p_{\infty, \mathcal{C}}^{(i)}(x) := \frac{1}{c_i} p_\infty(x) \mathbf{1}_{\mathcal{C}_i}(x), \quad p_{\infty, \mathcal{C}}^{(i)}(dx) := p_{\infty, \mathcal{C}}^{(i)}(x) dx.$$

Define the core means

$$\bar{f}_{\mathcal{C}_i} := \int_{\mathcal{C}_i} f(x) p_{\infty, \mathcal{C}}^{(i)}(dx).$$

Finally define the core fractions inside each region:

$$\eta_i := p_\infty^{(i)}(\mathcal{C}_i) = \frac{c_i}{w_i} \in (0, 1].$$

3.4.3 Dirichlet Forms and Spectral Gap

Diffusion and jump Dirichlet forms. Let $\sigma > 0$ denote the diffusion scale. Define

$$\mathcal{E}_{\text{diff}}(f, f) := \frac{\sigma^2}{2} \int_{\mathbb{R}^d} |\nabla f(x)|^2 p_\infty(x) dx,$$

and for a jump measure ν define

$$\mathcal{E}_{\text{jump}}(f, f) := \frac{1}{2} \int_{\mathbb{R}^d} p_\infty(x) \int_{\mathbb{R}^d} (f(x+r) - f(x))^2 \nu(dr) dx.$$

The total Dirichlet form is

$$\mathcal{E}_{\text{levy}}(f, f) := \mathcal{E}_{\text{diff}}(f, f) + \mathcal{E}_{\text{jump}}(f, f).$$

Spectral gap via Poincaré inequality. Define the spectral gap $\lambda(\mathcal{G})$ by the best constant in

$$\text{Var}_{p_\infty}(f) \leq \frac{1}{\lambda(\mathcal{G})} \mathcal{E}_{\text{levy}}(f, f).$$

3.4.4 Within-Well Mixing Rate λ_{in}

Restricted Dirichlet forms inside each well. Recall that the diffusion Dirichlet form is

$$\mathcal{E}_{\text{diff}}(f, f) = \frac{\sigma^2}{2} \int_{\mathbb{R}^d} |\nabla f(x)|^2 p_\infty(x) dx.$$

For each basin Ω_i we define the within-well diffusion Dirichlet form by restricting both the domain and the reference measure:

$$\mathcal{E}_{\text{diff}}^{(i)}(f, f) := \frac{\sigma^2}{2} \int_{\Omega_i} |\nabla f(x)|^2 p_\infty^{(i)}(dx), \quad i \in \{1, 2\}.$$

The corresponding within-well spectral gap is

$$\lambda_{\text{in}}^{(i)} := \inf \left\{ \frac{\mathcal{E}_{\text{diff}}^{(i)}(f, f)}{\text{Var}_{p_\infty^{(i)}}(f)} : f \in C_c^\infty(\Omega_i), \text{Var}_{p_\infty^{(i)}}(f) > 0 \right\},$$

that is, $\lambda_{\text{in}}^{(i)}$ is the best constant in the Poincaré inequality

$$\text{Var}_{p_{\infty}^{(i)}}(f) \leq \frac{1}{\lambda_{\text{in}}^{(i)}} \mathcal{E}_{\text{diff}}^{(i)}(f, f), \quad \forall f \in C_c^{\infty}(\Omega_i). \quad (42)$$

We then define the overall within-well mixing rate as

$$\lambda_{\text{in}} := \min\{\lambda_{\text{in}}^{(1)}, \lambda_{\text{in}}^{(2)}\}.$$

Strong convexity on the core sets. Recall the core sets

$$\mathcal{C}_i := B(m_i, r_c), \quad i \in \{1, 2\},$$

introduced in Section 3.4.2. We now assume that the potential is uniformly strongly convex on each core.

Assumption 3.13 (Core strong convexity). *There exists a constant $m_{\text{in}} > 0$ such that*

$$\nabla^2 V(x) \succeq m_{\text{in}} \text{Id}_d \quad \text{for all } x \in \mathcal{C}_1 \cup \mathcal{C}_2.$$

Under Assumption 3.13, the invariant density p_{∞} is locally strongly log-concave on each $\mathcal{C}_i$. More precisely, for the diffusion

$$dX_t = -\nabla V(X_t) dt + \sigma dW_t,$$

the invariant density is (up to normalization) $p_{\infty}(x) \propto \exp(-\frac{2}{\sigma^2} V(x))$. Thus on $\mathcal{C}_i$ we have

$$\nabla^2 \left(\frac{2}{\sigma^2} V(x) \right) \succeq \frac{2m_{\text{in}}}{\sigma^2} \text{Id}_d.$$

By the Brascamp–Lieb inequality, the core-conditional measures $p_{\infty, \mathcal{C}}^{(i)}$ satisfy a uniform local Poincaré inequality: for all smooth $g : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\text{Var}_{p_{\infty, \mathcal{C}}^{(i)}}(g) \leq \kappa_{\text{core}} \int_{\mathcal{C}_i} |\nabla g(x)|^2 p_{\infty, \mathcal{C}}^{(i)}(dx), \quad i \in \{1, 2\}, \quad (43)$$

where we can take the explicit bound

$$\kappa_{\text{core}} \leq \frac{\sigma^2}{2m_{\text{in}}}. \quad (44)$$

Lyapunov drift outside the cores. We now impose a coercivity condition on V outside each core, along rays emanating from the corresponding local minimum m_i . For $x \in \mathbb{R}^d$ we write

$$r_i(x) := |x - m_i|, \quad i \in \{1, 2\}.$$

Assumption 3.14 (Radial drift outside the core). *For each $i \in \{1, 2\}$ there exists a constant $\alpha_i > 0$ such that*

$$\langle x - m_i, \nabla V(x) \rangle \geq \alpha_i r_i(x) \quad \text{for all } x \in \Omega_i \text{ with } r_i(x) \geq r_c. \quad (45)$$

We now construct an explicit Lyapunov function W_i on Ω_i . Fix $\gamma_i > 0$ (to be chosen below) and set

$$W_i(x) := \begin{cases} \exp(\gamma_i r_i(x)), & r_i(x) \geq r_c, \\ \text{any } C^2 \text{ extension with } W_i(x) \geq 1, & r_i(x) < r_c. \end{cases}$$

By construction $W_i \in C^2(\Omega_i)$ and $W_i \geq 1$ everywhere. On the exterior region $\{x \in \Omega_i : r_i(x) \geq r_c\}$, we have

$$\nabla W_i(x) = \gamma_i \frac{x - m_i}{r_i(x)} W_i(x), \quad \Delta W_i(x) = \left(\gamma_i^2 + \gamma_i \frac{d-1}{r_i(x)} \right) W_i(x).$$

The diffusion generator reads

$$L_{\text{diff}} f(x) := \frac{\sigma^2}{2} \Delta f(x) - \nabla V(x) \cdot \nabla f(x),$$

and we compute, for $r_i(x) \geq r_c$,

$$\begin{aligned} L_{\text{diff}} W_i(x) &= \frac{\sigma^2}{2} \Delta W_i(x) - \nabla V(x) \cdot \nabla W_i(x) \\ &= \left[\frac{\sigma^2}{2} \left(\gamma_i^2 + \gamma_i \frac{d-1}{r_i(x)} \right) - \gamma_i \frac{\langle x - m_i, \nabla V(x) \rangle}{r_i(x)} \right] W_i(x). \end{aligned}$$

Using Assumption 3.16, we obtain for all $x \in \Omega_i$ with $r_i(x) \geq r_c$,

$$\begin{aligned} L_{\text{diff}} W_i(x) &\leq \left[\frac{\sigma^2}{2} \left(\gamma_i^2 + \gamma_i \frac{d-1}{r_i(x)} \right) - \gamma_i \alpha_i \right] W_i(x) \\ &\leq -\gamma_i \left[\alpha_i - \frac{\sigma^2}{2} \gamma_i - \frac{\sigma^2}{2} \frac{d-1}{r_c} \right] W_i(x). \end{aligned}$$

Assume that

$$\alpha_i > \frac{\sigma^2}{2} \frac{d-1}{r_c},$$

and choose $\gamma_i > 0$ such that

$$0 < \gamma_i < \alpha_i - \frac{\sigma^2}{2} \frac{d-1}{r_c}.$$

We then define the explicit drift constant

$$\theta_i := \gamma_i \left(\alpha_i - \frac{\sigma^2}{2} \gamma_i - \frac{\sigma^2}{2} \frac{d-1}{r_c} \right) > 0 \quad (46)$$

and conclude that

$$L_{\text{diff}} W_i(x) \leq -\theta_i W_i(x), \quad x \in \Omega_i, \quad r_i(x) \geq r_c. \quad (47)$$

On the core region $\mathcal{C}_i$ we define the finite constant

$$b_i := \sup_{x \in \mathcal{C}_i} \left(L_{\text{diff}} W_i(x) + \theta_i W_i(x) \right)_+, \quad (a)_+ := \max\{a, 0\}. \quad (48)$$

Since $\mathcal{C}_i$ is compact and $W_i \in C^2(\Omega_i)$, this supremum is finite. Combining (47) with (48) yields the global Lyapunov condition

$$L_{\text{diff}} W_i(x) \leq -\theta_i W_i(x) + b_i \mathbf{1}_{\mathcal{C}_i}(x), \quad \forall x \in \Omega_i. \quad (49)$$

Application of the BCG Lyapunov criterion. We now recall the relevant result of Bakry, Barthe, Cattiaux and Guillin (BCG) that links Lyapunov conditions and global Poincaré inequalities.

Theorem 3.15 (BCG Lyapunov criterion for Poincaré, informal statement). *Let μ be a probability measure on $\mathbb{R}^d$ which is reversible for a diffusion generator L with carré du champ Γ . Assume there exist a measurable function $W \geq 1$, constants $\theta > 0, b < \infty$, and a measurable set $K \subset \mathbb{R}^d$ such that*

$$LW \leq -\theta W + b \mathbf{1}_K \quad \text{on } \mathbb{R}^d,$$

and that the restriction of μ to K satisfies a local Poincaré inequality with constant $C_P(K)$. Then μ satisfies a global Poincaré inequality

$$\text{Var}_\mu(f) \leq \frac{1}{\theta} (1 + b C_P(K)) \int \Gamma(f, f) d\mu, \quad \forall f \in C_c^\infty(\mathbb{R}^d).$$

(See, for instance, Theorem 1.2 in Bakry–Barthe–Cattiaux–Guillin.)

We apply Theorem 3.15 to each conditional invariant measure $p_\infty^{(i)}$ with the following choices:

$$\begin{aligned} \mu &= p_\infty^{(i)}, & L &= L_{\text{diff}}, & \Gamma(f, f) &= \frac{\sigma^2}{2} |\nabla f|^2, \\ W &= W_i, & \theta &= \theta_i, & b &= b_i, & K &= \mathcal{C}_i. \end{aligned}$$

The Lyapunov condition (49) provides the required drift inequality, while the local Poincaré inequality on $K = \mathcal{C}_i$ is given by (43) with constant $C_P(K) = \kappa_{\text{core}}$.

Therefore, Theorem 3.15 yields, for each $i \in \{1, 2\}$ and every smooth $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\text{Var}_{p_\infty^{(i)}}(f) \leq \frac{1}{\theta_i} (1 + b_i \kappa_{\text{core}}) \int_{\Omega_i} \Gamma(f, f) p_\infty^{(i)}(dx) = \frac{\sigma^2}{2} \frac{1 + b_i \kappa_{\text{core}}}{\theta_i} \int_{\Omega_i} |\nabla f(x)|^2 p_\infty^{(i)}(dx).$$

Comparing with (52), we obtain the explicit lower bound on the within-well spectral gap

$$\lambda_{\text{in}}^{(i)} \geq \frac{\sigma^2 \theta_i}{2(1 + b_i \kappa_{\text{core}})}, \quad i \in \{1, 2\}, \quad (50)$$

where θ_i and b_i are given by (46) and (48), and κ_{core} satisfies (44).

Finally, since

$$\lambda_{\text{in}} := \min\{\lambda_{\text{in}}^{(1)}, \lambda_{\text{in}}^{(2)}\},$$

we arrive at the global within-well mixing bound

$$\lambda_{\text{in}} \geq \min_{i \in \{1, 2\}} \frac{\sigma^2 \theta_i}{2(1 + b_i \kappa_{\text{core}})}, \quad \kappa_{\text{core}} \leq \frac{\sigma^2}{2m_{\text{in}}}. \quad (51)$$

This shows that the within-well mixing rate is controlled explicitly by the core strong convexity parameter m_{in} , the radial drift constants α_i , the core radius r_c , and the scale σ of the diffusion.

3.4.5 Within-Well Mixing Rate λ_{in}

Within-well Dirichlet forms and spectral gaps. Recall that the diffusion Dirichlet form is

$$\mathcal{E}_{\text{diff}}(f, f) = \frac{\sigma^2}{2} \int_{\mathbb{R}^d} |\nabla f(x)|^2 p_{\infty}(x) dx.$$

For each basin Ω_i we define the within-well diffusion Dirichlet form by restricting both the domain and the reference measure:

$$\mathcal{E}_{\text{diff}}^{(i)}(f, f) := \frac{\sigma^2}{2} \int_{\Omega_i} |\nabla f(x)|^2 p_{\infty}^{(i)}(dx), \quad i \in \{1, 2\},$$

where $p_{\infty}^{(i)}$ is the normalized restriction of p_{∞} to Ω_i . The associated within-well spectral gap is

$$\lambda_{\text{in}}^{(i)} := \inf \left\{ \frac{\mathcal{E}_{\text{diff}}^{(i)}(f, f)}{\text{Var}_{p_{\infty}^{(i)}}(f)} : f \in C_c^{\infty}(\Omega_i), \text{Var}_{p_{\infty}^{(i)}}(f) > 0 \right\},$$

that is, $\lambda_{\text{in}}^{(i)}$ is the best constant in the Poincaré inequality

$$\text{Var}_{p_{\infty}^{(i)}}(f) \leq \frac{1}{\lambda_{\text{in}}^{(i)}} \mathcal{E}_{\text{diff}}^{(i)}(f, f), \quad \forall f \in C_c^{\infty}(\Omega_i). \quad (52)$$

We then define the overall within-well mixing rate as

$$\lambda_{\text{in}} := \min \{ \lambda_{\text{in}}^{(1)}, \lambda_{\text{in}}^{(2)} \}.$$

Radial drift outside the cores. For $x \in \mathbb{R}^d$ and each $i \in \{1, 2\}$, write

$$r_i(x) := |x - m_i|,$$

where m_i is the local minimum of V in Ω_i . We impose the following coercive drift condition along rays emanating from m_i .

Assumption 3.16 (Radial drift outside the core). *There exist constants $\alpha_i > 0$ and $r_c > 0$ such that, for each $i \in \{1, 2\}$,*

$$\langle x - m_i, \nabla V(x) \rangle \geq \alpha_i r_i(x) \quad \text{for all } x \in \Omega_i \text{ with } r_i(x) \geq r_c. \quad (53)$$

Note that we only assume $\alpha_i > 0$; no additional relation between α_i and r_c is required.

Enlarged balls and local Poincaré constants. Motivated by Assumption 3.16, we introduce a radius R_i and the corresponding ball K_i around m_i by

$$R_i := \max \left\{ r_c, \frac{2\sigma^2(d-1)}{\alpha_i} \right\}, \quad K_i := B(m_i, R_i), \quad i \in \{1, 2\}. \quad (54)$$

We denote by $p_{\infty, K}^{(i)}$ the conditional measure of $p_{\infty}^{(i)}$ on K_i , that is,

$$p_{\infty, K}^{(i)}(A) := \frac{p_{\infty}^{(i)}(A \cap K_i)}{p_{\infty}^{(i)}(K_i)}, \quad A \subset \mathbb{R}^d \text{ Borel}.$$

Since V is continuous and K_i is a compact ball, we have

$$V_{\min}^{(i)} := \inf_{x \in K_i} V(x) > -\infty, \quad V_{\max}^{(i)} := \sup_{x \in K_i} V(x) < +\infty.$$

Recall that the invariant density of the diffusion

$$dX_t = -\nabla V(X_t) dt + \sigma dW_t$$

is (up to normalization) given by $p_\infty(x) \propto \exp(-\frac{2}{\sigma^2} V(x))$. Hence, on K_i ,

$$\frac{\sup_{x \in K_i} p_\infty^{(i)}(x)}{\inf_{x \in K_i} p_\infty^{(i)}(x)} \leq \exp\left(\frac{2}{\sigma^2} (V_{\max}^{(i)} - V_{\min}^{(i)})\right). \quad (55)$$

It is classical that the Lebesgue measure restricted to the ball K_i satisfies a Poincaré inequality with constant of order R_i^2 . Combining this with the density bound (55), one obtains a local Poincaré inequality for $p_{\infty, K}^{(i)}$ on K_i with explicit constant; see for instance the discussion before Theorem 1.4 in Bakry–Barthe–Cattiaux–Guillin [45], where the authors derive the (non-optimal but explicit) bound $\kappa_R \lesssim R^2 \exp(\text{Osc}_R V)$ for μ restricted to a ball $B(0, R)$. Specialized to our potential $\tilde{V}(x) = \frac{2}{\sigma^2} V(x)$, this yields the following lemma.

Lemma 3.17 (Local Poincaré inequality on K_i). *For each $i \in \{1, 2\}$ and every smooth $g : \mathbb{R}^d \rightarrow \mathbb{R}$,*

$$\text{Var}_{p_{\infty, K}^{(i)}}(g) \leq C_{K_i} \int_{K_i} |\nabla g(x)|^2 p_{\infty, K}^{(i)}(dx), \quad (56)$$

where the local Poincaré constant can be chosen explicitly as

$$C_{K_i} \leq c_d R_i^2 \exp\left(\frac{2}{\sigma^2} (V_{\max}^{(i)} - V_{\min}^{(i)})\right), \quad (57)$$

for some finite constant $c_d > 0$ depending only on the dimension d , and R_i is given in (54).

Lyapunov function on each well. Fix $i \in \{1, 2\}$. In order to apply the Lyapunov criterion of Bakry–Barthe–Cattiaux–Guillin [45], we construct a radial Lyapunov function around m_i . Choose

$$\gamma_i := \frac{\alpha_i}{\sigma^2}, \quad (58)$$

and define

$$W_i(x) := \begin{cases} \exp(\gamma_i r_i(x)), & r_i(x) \geq R_i, \\ \text{any } C^2 \text{ extension with } W_i(x) \geq 1, & r_i(x) < R_i, \end{cases} \quad (59)$$

so that $W_i \in C^2(\Omega_i)$ and $W_i \geq 1$ everywhere.

On the region $\{x \in \Omega_i : r_i(x) \geq R_i\}$ we have

$$\nabla W_i(x) = \gamma_i \frac{x - m_i}{r_i(x)} W_i(x), \quad \Delta W_i(x) = \left(\gamma_i^2 + \gamma_i \frac{d-1}{r_i(x)}\right) W_i(x).$$

Recall that the diffusion generator is

$$L_{\text{diff}} f(x) := \frac{\sigma^2}{2} \Delta f(x) - \nabla V(x) \cdot \nabla f(x).$$

Using Assumption 3.16, we compute, for $r_i(x) \geq R_i$,

$$\begin{aligned}
L_{\text{diff}} W_i(x) &= \frac{\sigma^2}{2} \Delta W_i(x) - \nabla V(x) \cdot \nabla W_i(x) \\
&= \left[\frac{\sigma^2}{2} \left(\gamma_i^2 + \gamma_i \frac{d-1}{r_i(x)} \right) - \gamma_i \frac{\langle x - m_i, \nabla V(x) \rangle}{r_i(x)} \right] W_i(x) \\
&\leq \left[\frac{\sigma^2}{2} \left(\gamma_i^2 + \gamma_i \frac{d-1}{r_i(x)} \right) - \gamma_i \alpha_i \right] W_i(x) \\
&= -\gamma_i \left[\alpha_i - \frac{\sigma^2}{2} \gamma_i - \frac{\sigma^2}{2} \frac{d-1}{r_i(x)} \right] W_i(x).
\end{aligned} \tag{60}$$

With the choice $\gamma_i = \alpha_i / \sigma^2$, we have

$$\delta_i := \alpha_i - \frac{\sigma^2}{2} \gamma_i = \frac{\alpha_i}{2} > 0. \tag{61}$$

By the definition of R_i in (54), for all x with $r_i(x) \geq R_i$,

$$\delta_i - \frac{\sigma^2}{2} \frac{d-1}{r_i(x)} \geq \frac{\delta_i}{2} = \frac{\alpha_i}{4},$$

and therefore, combining this with (60),

$$L_{\text{diff}} W_i(x) \leq -\theta_i W_i(x), \quad x \in \Omega_i, \quad r_i(x) \geq R_i, \tag{62}$$

where we set

$$\theta_i := \frac{\gamma_i \delta_i}{2} = \frac{1}{2} \cdot \frac{\alpha_i}{\sigma^2} \cdot \frac{\alpha_i}{2} = \frac{\alpha_i^2}{4\sigma^2} > 0. \tag{63}$$

On the compact ball $K_i = B(m_i, R_i)$ we define

$$b_i := \sup_{x \in K_i} \left(L_{\text{diff}} W_i(x) + \theta_i W_i(x) \right)_+, \quad (a)_+ := \max\{a, 0\}. \tag{64}$$

Since K_i is compact and $W_i \in C^2(\Omega_i)$, this supremum is finite. Combining (62) with (64) yields the global Lyapunov condition

$$L_{\text{diff}} W_i(x) \leq -\theta_i W_i(x) + b_i \mathbf{1}_{K_i}(x), \quad \forall x \in \Omega_i. \tag{65}$$

Application of the BCG Lyapunov–Poincaré criterion. We now recall the following result, which is a localized version of Theorem 1.4 in Bakry–Barthe–Cattiaux–Guillin [45].

Theorem 3.18 (Lyapunov criterion for Poincaré). *Let μ be a probability measure on $\mathbb{R}^d$ which is reversible for a diffusion generator L with carré du champ Γ . Assume there exist a measurable function $W \geq 1$, constants $\theta > 0$, $b < \infty$, and a measurable set $K \subset \mathbb{R}^d$ such that*

$$LW \leq -\theta W + b \mathbf{1}_K \quad \text{on } \mathbb{R}^d,$$

and that the restriction μ_K of μ to K satisfies a local Poincaré inequality with constant C_K , i.e.

$$\text{Var}_{\mu_K}(g) \leq C_K \int_K \Gamma(g, g) d\mu_K, \quad \forall g \in C_c^\infty(\mathbb{R}^d).$$

Then μ satisfies a global Poincaré inequality

$$\text{Var}_\mu(f) \leq \frac{1}{\theta} (1 + b C_K) \int \Gamma(f, f) d\mu, \quad \forall f \in C_c^\infty(\mathbb{R}^d).$$

We apply Theorem 3.18 to each conditional invariant measure $p_\infty^{(i)}$ with the choices

$$\begin{aligned}\mu &= p_\infty^{(i)}, & L &= L_{\text{diff}}, & \Gamma(f, f) &= \frac{\sigma^2}{2} |\nabla f|^2, \\ W &= W_i, & \theta &= \theta_i, & b &= b_i, & K &= K_i.\end{aligned}$$

The Lyapunov condition (65) provides the drift inequality, while Lemma 3.17 provides the local Poincaré inequality on K_i with constant C_{K_i} .

Therefore, for each $i \in \{1, 2\}$ and every smooth $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\text{Var}_{p_\infty^{(i)}}(f) \leq \frac{1}{\theta_i} (1 + b_i C_{K_i}) \int_{\Omega_i} \frac{\sigma^2}{2} |\nabla f(x)|^2 p_\infty^{(i)}(dx).$$

Comparing with (52), and recalling that the Dirichlet form is $\mathcal{E}_{\text{diff}}^{(i)}(f, f) = \frac{\sigma^2}{2} \int_{\Omega_i} |\nabla f|^2 p_\infty^{(i)}(dx)$, we obtain

$$\lambda_{\text{in}}^{(i)} \geq \frac{\sigma^2 \theta_i}{2(1 + b_i C_{K_i})}. \quad (66)$$

Using the explicit value $\theta_i = \alpha_i^2 / (4\sigma^2)$ from (63), this becomes

$$\lambda_{\text{in}}^{(i)} \geq \frac{\alpha_i^2}{8(1 + b_i C_{K_i})}. \quad (67)$$

Finally, plugging the bound (57) on C_{K_i} , we obtain the fully explicit estimate

$$\lambda_{\text{in}}^{(i)} \geq \frac{\alpha_i^2}{8 \left(1 + b_i c_d R_i^2 \exp\left(\frac{2}{\sigma^2} (V_{\text{max}}^{(i)} - V_{\text{min}}^{(i)})\right) \right)}, \quad R_i = \max \left\{ r_c, \frac{2\sigma^2(d-1)}{\alpha_i} \right\}. \quad (68)$$

Since

$$\lambda_{\text{in}} := \min \{ \lambda_{\text{in}}^{(1)}, \lambda_{\text{in}}^{(2)} \},$$

we conclude that

$$\lambda_{\text{in}} \geq \min_{i \in \{1, 2\}} \frac{\alpha_i^2}{8 \left(1 + b_i c_d R_i^2 \exp\left(\frac{2}{\sigma^2} (V_{\text{max}}^{(i)} - V_{\text{min}}^{(i)})\right) \right)}, \quad R_i = \max \left\{ r_c, \frac{2\sigma^2(d-1)}{\alpha_i} \right\}. \quad (69)$$

This gives an explicit lower bound on the within-well mixing rate in terms of the drift constants α_i , the core radius r_c , the enlarged radii R_i , the local oscillations $V_{\text{max}}^{(i)} - V_{\text{min}}^{(i)}$, the noise level σ , and the dimension-dependent constant c_d , without any convexity assumption on V .

3.4.6 Controlling Inter-Well Variance: Barrier-Free Coupling via Global Jumps

In the variance decomposition

$$\text{Var}_{p_\infty}(f) = \sum_{i=1}^2 w_i \text{Var}_{p_\infty^{(i)}}(f) + w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2, \quad (70)$$

the first term describes fluctuations *within* each well, while the second term

$$w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2$$

measures the difference between the mean values in the two wells. For a pure diffusion, reducing this term requires transferring mass across the saddle, so classical bounds involve the barrier height. In this section we show that, in the presence of a global jump component, one can *directly jump from one core to the other*, and obtain an inter-well constant λ_{inter} which does *not* depend on the saddle height.

Our goal is an inequality of the form

$$w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2 \leq \frac{C_{\text{in}}}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f) + \frac{1}{\lambda_{\text{inter}}} \mathcal{E}_{\text{jump}}(f, f), \quad (71)$$

where the first term on the right-hand side depends only on within-well diffusion (no jumps are needed), and the second term depends only on the global jump part. We will derive an explicit lower bound on λ_{inter} in terms of λ_0 , the shell density lower bound $g_\star$, the geometric parameters D, r_c , and some local core constants; in particular, no saddle information enters λ_{inter} .

Global jump structure and shell density lower bound. Recall our assumption that the jump measure contains a global component

$$\nu_{\text{glob}}(\text{d}r) = \lambda_0 \nu_J(\text{d}r), \quad \lambda_0 > 0,$$

where ν_J is a probability measure on $\mathbb{R}^d$ with isotropic density

$$\nu_J(\text{d}r) = j(r) \text{d}r, \quad j(r) = j_0(|r|).$$

Let $R \sim \nu_J$ and $S := |R|$. The radial density g of S is given by

$$g(s) = |\mathbb{S}^{d-1}| s^{d-1} j_0(s), \quad s > 0,$$

where $|\mathbb{S}^{d-1}| = d\omega_d$ and $\omega_d := \text{Vol}(B(0, 1))$.

Let $D := |m_2 - m_1|$ be the distance between the two minima and $r_c > 0$ the core radius. We assume a uniform lower bound on the radial density on a shell around D : there exists $g_\star > 0$ such that

$$g(s) \geq g_\star \quad \forall s \in [D - 2r_c, D + 2r_c]. \quad (72)$$

Equivalently, since $g(s) = |\mathbb{S}^{d-1}| s^{d-1} j_0(s)$, for all $s \in [D - 2r_c, D + 2r_c]$ we have

$$j_0(s) \geq \frac{g_\star}{|\mathbb{S}^{d-1}| (D + 2r_c)^{d-1}} = \frac{g_\star}{d\omega_d (D + 2r_c)^{d-1}} =: j_\star. \quad (73)$$

Geometrically, for any $x \in \mathcal{C}_1, y \in \mathcal{C}_2$,

$$|y - x| \in [D - 2r_c, D + 2r_c],$$

hence by (73) we have $j_0(|y - x|) \geq j_\star$.

A basic Jensen inequality: controlling core mean differences by a cross-core integral.

Recall that Ω_i is the basin of the i -th well, $\mathcal{C}_i \subset \Omega_i$ is the corresponding core, and

$$\begin{aligned} w_i &:= p_\infty(\Omega_i), & c_i &:= p_\infty(\mathcal{C}_i), \\ p_\infty^{(i)} &:= \frac{1}{w_i} p_\infty(\cdot \cap \Omega_i), & p_{\infty, \mathcal{C}}^{(i)} &:= \frac{1}{c_i} p_\infty(\cdot \cap \mathcal{C}_i). \end{aligned}$$

We write

$$\bar{f}_{\Omega_i} := \int_{\Omega_i} f \, dp_{\infty}^{(i)}, \quad \bar{f}_{\mathcal{C}_i} := \int_{\mathcal{C}_i} f \, dp_{\infty, \mathcal{C}}^{(i)}.$$

Let

$$X \sim p_{\infty, \mathcal{C}}^{(1)}, \quad Y \sim p_{\infty, \mathcal{C}}^{(2)}$$

be independent. Then

$$\mathbb{E}[f(X)] = \bar{f}_{\mathcal{C}_1}, \quad \mathbb{E}[f(Y)] = \bar{f}_{\mathcal{C}_2}.$$

Using $\text{Var}(f(Y) - f(X)) \geq 0$, or equivalently Jensen's inequality for the convex function $z \mapsto z^2$, we obtain

$$\mathbb{E}[(f(Y) - f(X))^2] \geq (\mathbb{E}[f(Y)] - \mathbb{E}[f(X)])^2 = (\bar{f}_{\mathcal{C}_2} - \bar{f}_{\mathcal{C}_1})^2.$$

Writing the expectation as an integral, this reads

$$\iint_{\mathcal{C}_1 \times \mathcal{C}_2} (f(y) - f(x))^2 p_{\infty, \mathcal{C}}^{(1)}(dx) p_{\infty, \mathcal{C}}^{(2)}(dy) \geq (\bar{f}_{\mathcal{C}_2} - \bar{f}_{\mathcal{C}_1})^2. \quad (74)$$

Using

$$p_{\infty, \mathcal{C}}^{(1)}(dx) = \frac{1}{c_1} p_{\infty}(x) \mathbf{1}_{\mathcal{C}_1}(x) \, dx, \quad p_{\infty, \mathcal{C}}^{(2)}(dy) = \frac{1}{c_2} p_{\infty}(y) \mathbf{1}_{\mathcal{C}_2}(y) \, dy,$$

we can rewrite (74) as

$$\iint_{\mathcal{C}_1 \times \mathcal{C}_2} (f(y) - f(x))^2 p_{\infty}(x) p_{\infty}(y) \, dx \, dy \geq c_1 c_2 (\bar{f}_{\mathcal{C}_2} - \bar{f}_{\mathcal{C}_1})^2. \quad (75)$$

This purely measure-theoretic inequality does not use the jump structure or the potential landscape. It shows that if we manage to control the cross-core integral on the left by the jump Dirichlet form, then we can control the difference between the core means.

Embedding the cross-core integral into the jump Dirichlet form via the shell lower bound. Recall the jump Dirichlet form

$$\mathcal{E}_{\text{jump}}(f, f) = \frac{1}{2} \int_{\mathbb{R}^d} p_{\infty}(x) \int_{\mathbb{R}^d} (f(x+r) - f(x))^2 \nu(dr) \, dx,$$

where ν contains the global component $\nu_{\text{glob}}(dr) = \lambda_0 \nu_J(dr)$. Let $\mathcal{E}_{\text{glob}}$ denote the contribution of ν_{glob} , so that

$$\mathcal{E}_{\text{jump}}(f, f) \geq \mathcal{E}_{\text{glob}}(f, f).$$

By the change of variables $y = x + r$ and using the density of ν_J , we get

$$\mathcal{E}_{\text{glob}}(f, f) = \frac{1}{2} \int_{\mathbb{R}^d} p_{\infty}(x) \int_{\mathbb{R}^d} (f(y) - f(x))^2 \lambda_0 j_0(|y - x|) \, dy \, dx.$$

Keeping only the contribution from jumps that start in $\mathcal{C}_1$ and land in $\mathcal{C}_2$ (discarding other non-negative terms lowers the right-hand side and hence yields a lower bound), we obtain

$$\mathcal{E}_{\text{glob}}(f, f) \geq \frac{1}{2} \int_{\mathcal{C}_1} p_{\infty}(x) \int_{\mathcal{C}_2} (f(y) - f(x))^2 \lambda_0 j_0(|y - x|) \, dy \, dx. \quad (76)$$

For $x \in \mathcal{C}_1$, $y \in \mathcal{C}_2$ we know $|y - x| \in [D - 2r_c, D + 2r_c]$, hence by (73), $j_0(|y - x|) \geq j_{\star}$, and therefore

$$\mathcal{E}_{\text{glob}}(f, f) \geq \frac{1}{2} \lambda_0 j_{\star} \int_{\mathcal{C}_1} p_{\infty}(x) \int_{\mathcal{C}_2} (f(y) - f(x))^2 \, dy \, dx.$$

We now convert the inner dy into a weighted measure $p_\infty(y) dy$ so that we can compare with (75). Introduce the local core supremum

$$M_C := \max \left\{ \sup_{x \in \mathcal{C}_1} p_\infty(x), \sup_{y \in \mathcal{C}_2} p_\infty(y) \right\}.$$

Since p_∞ is continuous and $\mathcal{C}_i$ are compact, $M_C < \infty$. For all $y \in \mathcal{C}_2$, $p_\infty(y) \leq M_C$, hence

$$dy \geq \frac{1}{M_C} p_\infty(y) dy.$$

Plugging this into the previous bound yields

$$\begin{aligned} \mathcal{E}_{\text{glob}}(f, f) &\geq \frac{1}{2} \lambda_0 j_\star \int_{\mathcal{C}_1} p_\infty(x) \int_{\mathcal{C}_2} (f(y) - f(x))^2 \frac{1}{M_C} p_\infty(y) dy dx \\ &= \frac{1}{2} \lambda_0 j_\star \frac{1}{M_C} \iint_{\mathcal{C}_1 \times \mathcal{C}_2} (f(y) - f(x))^2 p_\infty(x) p_\infty(y) dx dy. \end{aligned} \quad (77)$$

Combining (77) with (75), we obtain

$$c_1 c_2 (\bar{f}_{\mathcal{C}_2} - \bar{f}_{\mathcal{C}_1})^2 \leq \frac{2M_C}{\lambda_0 j_\star} \mathcal{E}_{\text{glob}}(f, f) \leq \frac{2M_C}{\lambda_0 j_\star} \mathcal{E}_{\text{jump}}(f, f). \quad (78)$$

From core means back to well means (purely within-well; no jumps needed). The previous step shows that the core mean difference $\bar{f}_{\mathcal{C}_1} - \bar{f}_{\mathcal{C}_2}$ is controlled by the jump energy. To connect this to the well mean difference $\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2}$ appearing in (70), observe that

$$\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2} = (\bar{f}_{\Omega_1} - \bar{f}_{\mathcal{C}_1}) + (\bar{f}_{\mathcal{C}_1} - \bar{f}_{\mathcal{C}_2}) + (\bar{f}_{\mathcal{C}_2} - \bar{f}_{\Omega_2}),$$

so by $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$,

$$(\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2 \leq 3 \left[(\bar{f}_{\Omega_1} - \bar{f}_{\mathcal{C}_1})^2 + (\bar{f}_{\mathcal{C}_1} - \bar{f}_{\mathcal{C}_2})^2 + (\bar{f}_{\mathcal{C}_2} - \bar{f}_{\Omega_2})^2 \right]. \quad (79)$$

We next estimate $\bar{f}_{\Omega_i} - \bar{f}_{\mathcal{C}_i}$ using within-well variance. Let $\eta_i := p_\infty^{(i)}(\mathcal{C}_i) = c_i/w_i$ be the mass of the core in the i -th well. A simple estimate (e.g. by Cauchy–Schwarz or a one-point Poincaré inequality) gives

$$(\bar{f}_{\Omega_i} - \bar{f}_{\mathcal{C}_i})^2 \leq \frac{1}{\eta_i} \text{Var}_{p_\infty^{(i)}}(f), \quad i = 1, 2. \quad (80)$$

Insert (80) into (79), multiply both sides by $w_1 w_2$, and recall that $w_1 w_2 = (c_1 c_2)/(\eta_1 \eta_2)$. Using (78), we obtain

$$\begin{aligned} w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2 &\leq 3 w_1 w_2 \left[\frac{1}{\eta_1} \text{Var}_{p_\infty^{(1)}}(f) + \frac{1}{\eta_2} \text{Var}_{p_\infty^{(2)}}(f) \right] + 3 w_1 w_2 (\bar{f}_{\mathcal{C}_1} - \bar{f}_{\mathcal{C}_2})^2 \\ &\leq 3 \left(\frac{w_1 w_2}{\eta_1} + \frac{w_1 w_2}{\eta_2} \right) \sum_{i=1}^2 \text{Var}_{p_\infty^{(i)}}(f) + \frac{6M_C}{\lambda_0 j_\star} \mathcal{E}_{\text{jump}}(f, f). \end{aligned} \quad (81)$$

Define

$$C_{\text{in}} := 3 \left(\frac{w_1 w_2}{\eta_1} + \frac{w_1 w_2}{\eta_2} \right). \quad (82)$$

On the other hand, from the within-well analysis we already have a lower bound on the within-well mixing rate λ_{in} such that

$$\sum_{i=1}^2 w_i \text{Var}_{p_{\infty}^{(i)}}(f) \leq \frac{1}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f). \quad (83)$$

Since all weights w_i, η_i are fixed positive constants determined by the equilibrium measure and the choice of cores, there exists a finite constant $C_{\text{in}} > 0$ (which can be chosen as in (82) up to a universal factor) such that

$$\sum_{i=1}^2 \text{Var}_{p_{\infty}^{(i)}}(f) \leq \frac{C_{\text{in}}}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f).$$

Plugging this into (81) and enlarging C_{in} if necessary, we obtain

$$w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2 \leq \frac{C_{\text{in}}}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f) + \frac{1}{\lambda_{\text{inter}}} \mathcal{E}_{\text{jump}}(f, f), \quad (84)$$

where

$$\lambda_{\text{inter}} := \frac{\lambda_0 j_{\star}}{6M_{\mathcal{C}}} \eta_1 \eta_2. \quad (85)$$

Structurally, λ_{inter} depends only on $\lambda_0, j_{\star}$ (hence on $g_{\star}, D, r_c$), and the local quantities $M_{\mathcal{C}}, \eta_1, \eta_2$; *it does not involve the saddle position or the barrier height.*

Explicit expression for λ_{inter} in terms of the shell lower bound $g_{\star}$. From (73) we have

$$j_{\star} = \frac{g_{\star}}{d \omega_d (D + 2r_c)^{d-1}}.$$

Substituting this into (85) yields

$$\lambda_{\text{inter}} = \frac{\lambda_0}{6M_{\mathcal{C}}} \eta_1 \eta_2 \frac{g_{\star}}{d \omega_d (D + 2r_c)^{d-1}}. \quad (86)$$

Thus, small values of λ_{inter} can arise from:

- The geometric “angular penalty” factor $(D + 2r_c)^{-(d-1)}$, reflecting the small solid angle occupied by the target core in high dimensions;
- A small shell density lower bound $g_{\star}$, i.e. if the jump distribution places little radial mass near radius D .

Crucially, there is no exponential suppression in terms of the potential barrier height; the bottleneck for inter-well motion is entirely controlled by jump geometry and shell density.

3.4.7 Combining Within-Well and Inter-Well Bounds: A Two-Scale Poincaré Inequality

We now combine the within-well bound (83) with the inter-well bound (84) in the variance decomposition (70). We obtain

$$\begin{aligned} \text{Var}_{p_{\infty}}(f) &= \sum_{i=1}^2 w_i \text{Var}_{p_{\infty}^{(i)}}(f) + w_1 w_2 (\bar{f}_{\Omega_1} - \bar{f}_{\Omega_2})^2 \\ &\leq \frac{1}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f) + \frac{C_{\text{in}}}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f) + \frac{1}{\lambda_{\text{inter}}} \mathcal{E}_{\text{jump}}(f, f) \\ &= \frac{1 + C_{\text{in}}}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f) + \frac{1}{\lambda_{\text{inter}}} \mathcal{E}_{\text{jump}}(f, f). \end{aligned}$$

If we redefine $\tilde{\lambda}_{\text{in}} := \lambda_{\text{in}}/(1 + C_{\text{in}})$ and, for simplicity of notation, relabel $\tilde{\lambda}_{\text{in}}$ again as λ_{in} (this only changes the constant by a fixed factor), we may rewrite the inequality as

$$\text{Var}_{p_\infty}(f) \leq \frac{1}{\lambda_{\text{in}}} \mathcal{E}_{\text{diff}}(f, f) + \frac{1}{\lambda_{\text{inter}}} \mathcal{E}_{\text{jump}}(f, f).$$

Since $\mathcal{E}_{\text{diff}}(f, f) \leq \mathcal{E}_{\text{levy}}(f, f)$ and $\mathcal{E}_{\text{jump}}(f, f) \leq \mathcal{E}_{\text{levy}}(f, f)$ (because $\mathcal{E}_{\text{levy}} = \mathcal{E}_{\text{diff}} + \mathcal{E}_{\text{jump}}$), we arrive at a two-scale Poincaré inequality

$$\text{Var}_{p_\infty}(f) \leq \left(\frac{1}{\lambda_{\text{in}}} + \frac{1}{\lambda_{\text{inter}}} \right) \mathcal{E}_{\text{levy}}(f, f). \quad (87)$$

This implies the following lower bound on the spectral gap $\lambda(\mathcal{G})$ of the generator $\mathcal{G}$:

$$\lambda(\mathcal{G}) \geq \left(\lambda_{\text{in}}^{-1} + \lambda_{\text{inter}}^{-1} \right)^{-1}. \quad (88)$$

3.4.8 Harmonic Mean Form: The Overall Rate is Governed by the Slower Scale (up to a Factor $\frac{1}{2}$)

For any $a, b > 0$,

$$(a^{-1} + b^{-1})^{-1} = \frac{1}{\frac{1}{a} + \frac{1}{b}} = \frac{ab}{a + b}. \quad (89)$$

Since $a + b \leq 2 \max\{a, b\}$, we deduce

$$\frac{ab}{a + b} \geq \frac{ab}{2 \max\{a, b\}} = \frac{1}{2} \min\{a, b\},$$

that is,

$$(a^{-1} + b^{-1})^{-1} \geq \frac{1}{2} \min\{a, b\}. \quad (90)$$

Applying (90) with $a = \lambda_{\text{in}}$, $b = \lambda_{\text{inter}}$ and combining with (88), we obtain the more “intuitive” bound

$$\lambda(\mathcal{G}) \geq \left(\lambda_{\text{in}}^{-1} + \lambda_{\text{inter}}^{-1} \right)^{-1} \geq \frac{1}{2} \min\{\lambda_{\text{in}}, \lambda_{\text{inter}}\}. \quad (91)$$

This shows that, up to a factor 1/2, the global convergence rate is governed by the slower of the two mechanisms:

- If within-well mixing is slow (λ_{in} small), then even very strong jumps cannot improve the overall spectral gap beyond this scale;
- If within-well mixing is fast but inter-well jumps are too weak (λ_{inter} small), then mass cannot be exchanged efficiently between wells, and the overall convergence remains slow.

3.4.9 Relation to the Saddle: What “Barrier-Free” Means Here

Along this line of argument we have derived two key constants:

- λ_{in} , the within-well spectral gap, controlled purely by local information near the minima (e.g. Hessian lower bounds or Lyapunov conditions);
- λ_{inter} , the inter-well rate generated by global jumps, depending only on the jump intensity λ_0 , the radial shell density lower bound $g_\star$, the geometry (D, r_c) , and the local core constants $M_{\mathcal{C}}, \eta_1, \eta_2$.

At no point did we use the saddle location or the barrier height. In particular, the expression (86) contains no factor of the form $\exp(-H/\sigma^2)$. Instead, the inter-well bottleneck is entirely controlled by the probability that a global jump lands in the other core, encoded by shell geometry and the density lower bound $g_\star$.

In words:

For a pure diffusion, inter-well moves must follow the energy landscape and typically go through the saddle, so the slowest step involves climbing the barrier and the spectral gap is often exponentially small in the barrier height.

With a suitably chosen global jump component, inter-well transitions no longer have to respect the potential landscape: they are governed by the jump distribution at radius $\approx D$, i.e. by the shell density and geometry. The metastable “barrier” is effectively bypassed, and the inter-well rate λ_{inter} is *barrier-free* in the sense that it is independent of the saddle height.

3.4.10 Two-Scale Poincaré Inequality and Harmonic-Mean Lower Bound

Combining the exact variance decomposition (41), the within-well bound (??), and the inter-well bound encoded in (85), we obtain a two-scale Poincaré inequality of the schematic form

$$\text{Var}_{p_\infty}(f) \leq \left(\frac{1}{\lambda_{\text{in}}} + \frac{1}{\lambda_{\text{inter}}} \right) \mathcal{E}_{\text{levy}}(f, f),$$

and hence the spectral gap satisfies

$$\lambda(\mathcal{G}) \geq \left(\lambda_{\text{in}}^{-1} + \lambda_{\text{inter}}^{-1} \right)^{-1}. \quad (92)$$

Finally, using the elementary inequality

$$(a^{-1} + b^{-1})^{-1} = \frac{ab}{a+b} \geq \frac{1}{2} \min\{a, b\},$$

we conclude the intuitive bound

$$\lambda(\mathcal{G}) \geq \frac{1}{2} \min\{\lambda_{\text{in}}, \lambda_{\text{inter}}\}. \quad (93)$$

4 algorithm

4.1 Lévy-score-based Monte Carlo

By virtue of Theorems 3.4, 3.7 and ??, the jump-diffusion SDE (25) provides a theoretically rigorous framework for sampling the Boltzmann distribution associated with (3). As a preliminary step, an appropriate Lévy process and its corresponding Lévy measure ν must be specified. In practice, this jump process must be amenable to efficient numerical implementation. One of the most computationally tractable choices is the compound Poisson process [38]:

$$L_t^{\text{cp}} = \sum_{i=1}^{N_t} A_i \delta(t - t_i), \quad (94)$$

where N_t denotes a Poisson process with intensity parameter λ_0 , and the jump sizes A_i are independent and identically distributed (i.i.d.) random variables drawn from a specified distribution

Algorithm 1 Lévy-score-based Monte Carlo sampling Boltzmann distribution

- 1: **Input:** An initial state $Z_{t_0} = z_0$, time step size Δt , total simulation time T , Poisson intensity parameter λ_0 , and an appropriate jump size distribution ν_J .
- 2: **for** $k = 0, 1, \dots, T/\Delta t - 1$ **do**
- 3: Sample the standard normal variable $\xi_k \sim \mathcal{N}(0, 1)$;
- 4: Sample the number of jumps $N_{\Delta t_k} \sim \text{Po}(\lambda_0 \Delta t)$, and sample the jump sizes $A_i \sim \nu_J$ for $i = 1, \dots, N_{\Delta t_k}$;
- 5: Propagate the system state:

$$Z_{t_{k+1}} = Z_{t_k} + \Delta t \left[-\nabla V(Z_{t_k}) - \int_0^1 \int_{\mathbb{R}^d - \{0\}} r \exp \left\{ -\frac{2V(Z_{t_k} - \theta r)}{\sigma^2} + \frac{2V(Z_{t_k})}{\sigma^2} \right\} \nu(dr) d\theta \right] \\ + \sigma \sqrt{\Delta t} \xi_k + \sum_{i=1}^{N_{\Delta t_k}} A_i;$$

- 6: Update the time variable $t_{k+1} = t_k + \Delta t$;
 - 7: **end for**
 - 8: **Output:** The trajectory samples $\{Z_{t_k}\}_{k=1}^{T/\Delta t}$ and the empirical distribution $p_N(x) := \frac{1}{T} \sum_{k=0}^{T/\Delta t} \delta(x - Z_{t_k})$.
-

ν_J . The parameter λ_0 governs the frequency of jump events, while ν_J dictates the distribution of the jump magnitudes. In this work, we adopt the compound Poisson process as the primary jump mechanism for the proposed sampling algorithm, see Algorithm 1.

Remark 4.1. *Determining the optimal choices for the Poisson intensity parameter λ_0 and the jump size distribution ν_J to maximize the efficiency of sampling the Boltzmann distribution p_∞ in (5) remains an open and pertinent question. The optimal configuration is inherently dependent on the specific landscape of the potential V . Because the energy barriers separating local minima profoundly limit the efficiency of standard Monte Carlo algorithms, ν_J should ideally be tailored such that the spatial jump magnitudes are sufficient to traverse the distances between distinct wells. If the typical jump sizes are too small to bypass these regional basins of attraction, the non-local process merely introduces localized fluctuations analogous to an auxiliary continuous diffusion, failing to facilitate the desired macroscopic state transitions. A rigorous theoretical and numerical investigation of these optimal parameter selections is reserved for future work.*

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